

CS 305: Computer Networks

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Link Layer

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Link layer, LANs: outline

6.1 introduction, services

6.2 error detection, correction

6.3 multiple access protocols

6.4 LANs

- addressing, ARP
- Ethernet
- switches
- VLANs

6.5 link virtualization: MPLS

6.6 data center networking

6.7 a day in the life of a web request

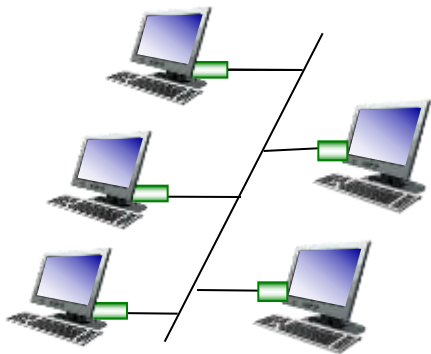
Ethernet

“dominant” wired LAN technology:

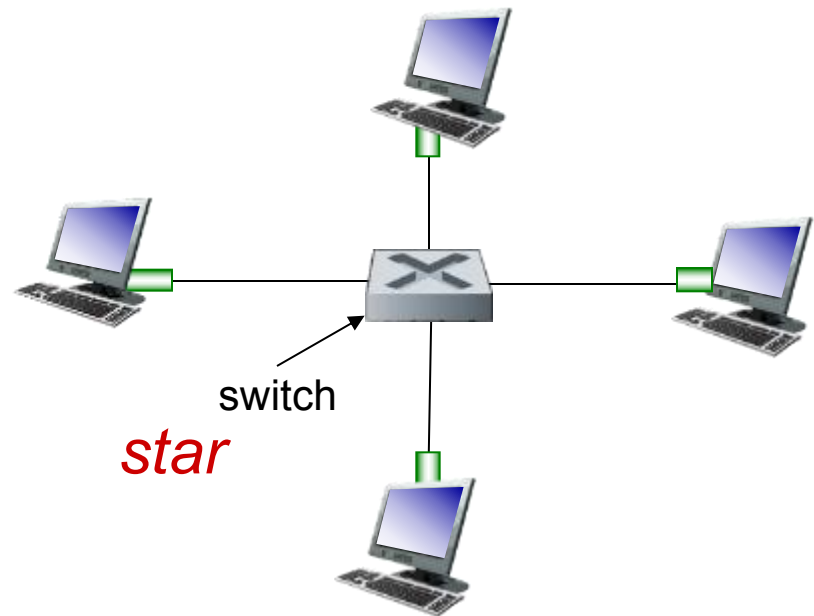
- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 10 Gbps

Ethernet: physical topology

- **bus:** popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- **star:** prevails today
 - active **switch** in center
 - each “spoke” runs a (separate) Ethernet protocol (nodes do not collide with each other)

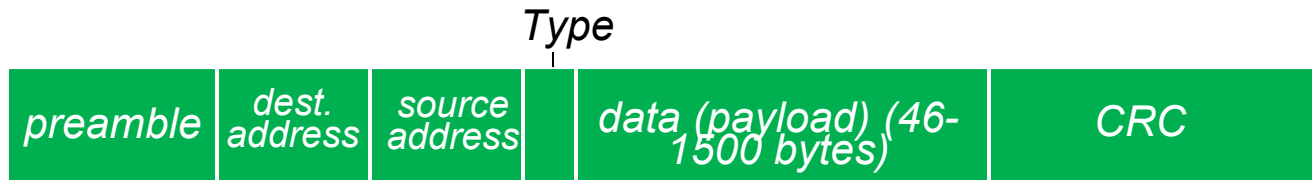


bus: coaxial cable



Ethernet frame structure

sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



preamble:

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- Transmit frame at 10Mbps, 100Mbps, 1Gbps
- Drift from target rate
- used to synchronize receiver, sender clock rates

Ethernet frame structure (more)

- *addresses*: 6 byte source, destination MAC addresses
 - if adapter receives frame with matching destination address, or with broadcast address (e.g. ARP packet), it passes data in frame to network layer protocol
 - otherwise, adapter discards frame
- *type*: indicates higher layer protocol (mostly IP but others possible, e.g., Novell IPX, AppleTalk)
- *CRC*: cyclic redundancy check at receiver
 - error detected: frame is dropped'

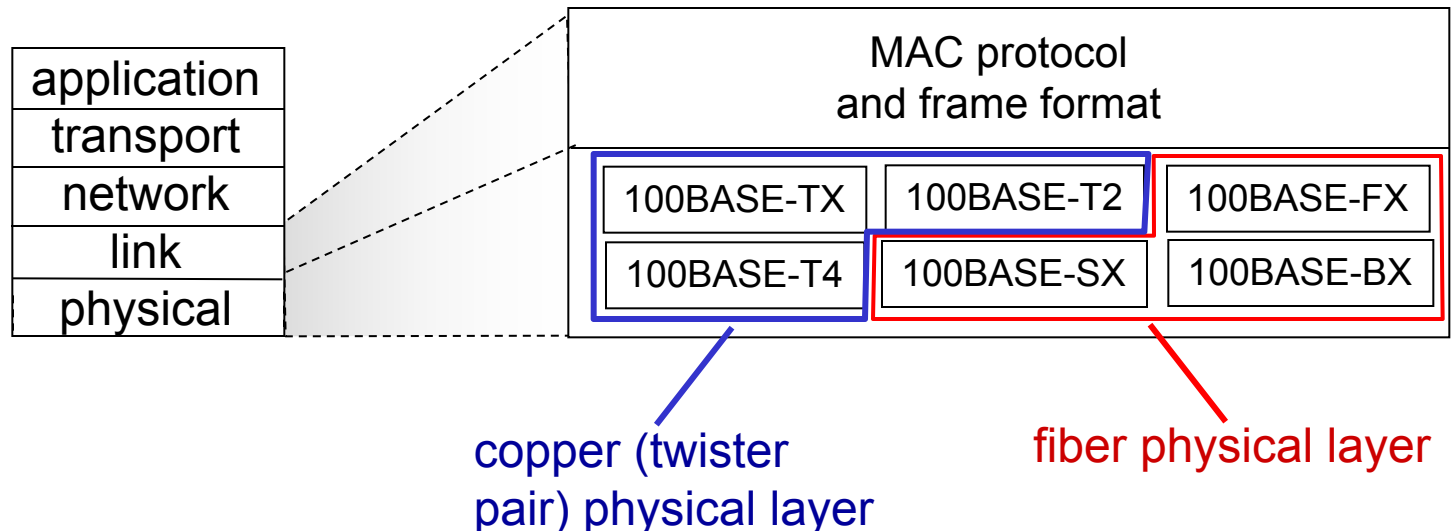


Ethernet: unreliable, connectionless

- *connectionless*: no handshaking between sending and receiving NICs
- *unreliable*: receiving NIC doesn't send acks or nacks to sending NIC
 - data in dropped frames recovered only if initial sender uses higher layer rdt (e.g., TCP), otherwise dropped data lost
 - Unaware of whether it is transmitting a brand-new datagram with brand-new data
- Ethernet's MAC protocol: unslotted *CSMA/CD with binary backoff*

802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards
 - common MAC protocol and frame format
 - different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 40 Gbps
 - different physical layer media: fiber, cable



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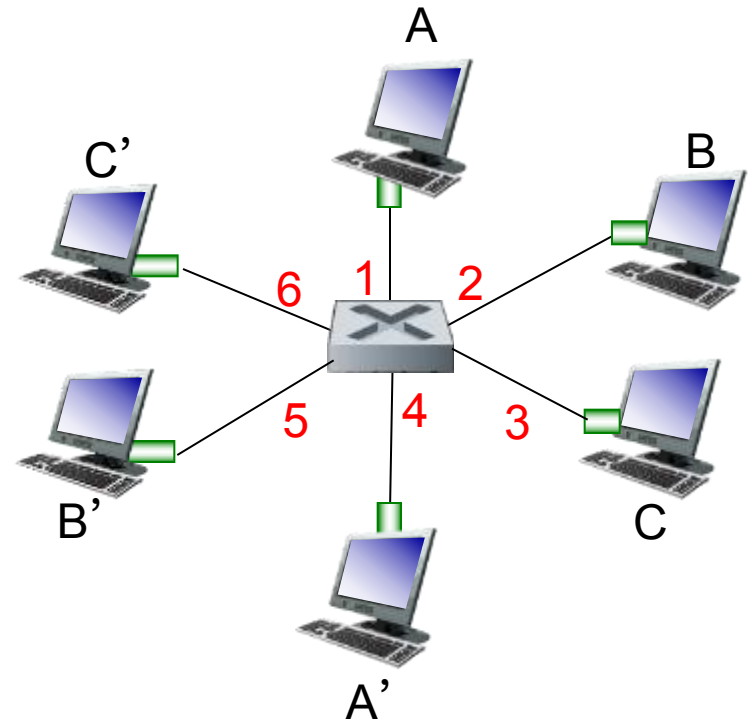
6.7 a day in the life of a web request

Ethernet switch

- **link-layer device**
 - store, forward Ethernet frames
 - examine incoming frame's MAC address, **selectively** forward frame to one-or-more outgoing links when frame is to be forwarded on segment
- *transparent*
 - hosts are unaware of presence of switches
- *plug-and-play, self-learning*
 - switches do not need to be configured

Switch: *multiple* simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
- *switching*: A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces
(1,2,3,4,5,6)

Switch forwarding table

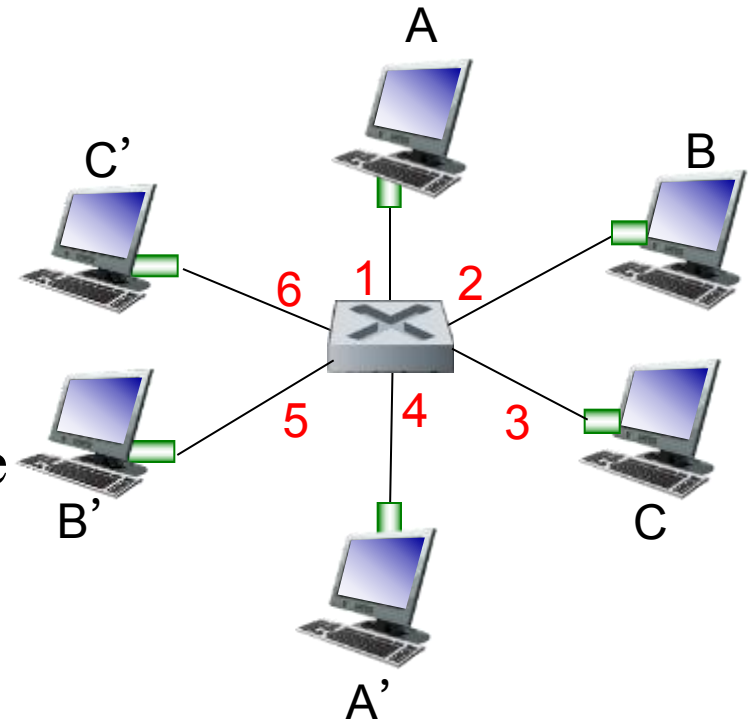
Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

A: each switch has a **switch table**, each entry:

- (MAC address of host, interface to reach host, time stamp)
- looks like a routing table!

Q: how are entries created, maintained in switch table?

- something like a routing protocol?

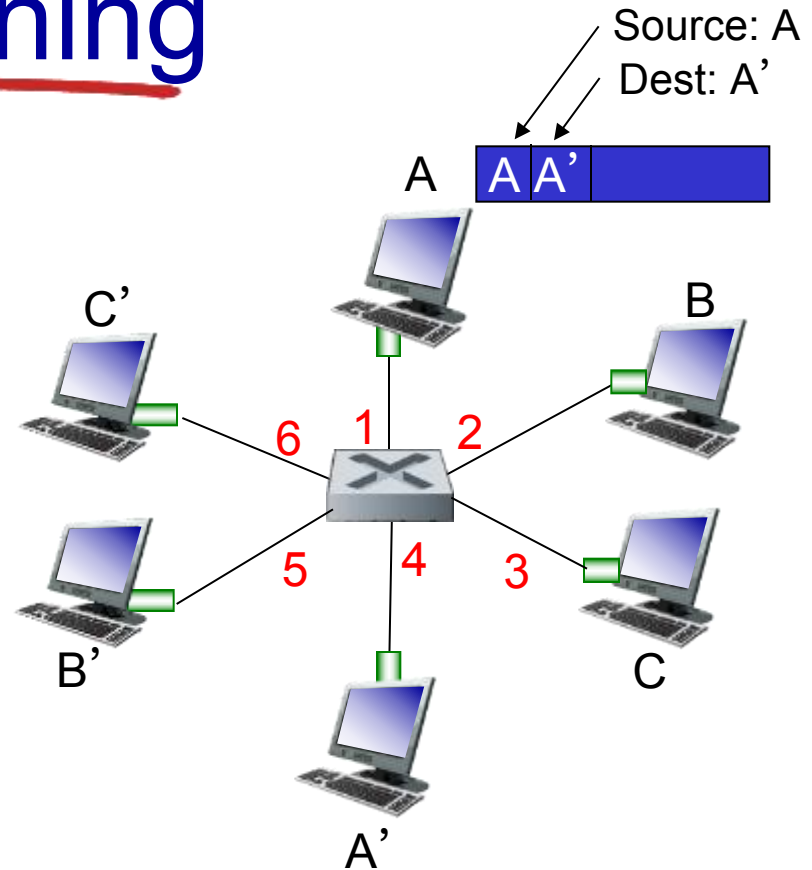


*switch with six interfaces
(1,2,3,4,5,6)*

Switch: self-learning

Switch *learns* which hosts can be reached through which interfaces

- when frame received, switch “learns” location of sender: incoming LAN segment
- records sender/location pair in switch table

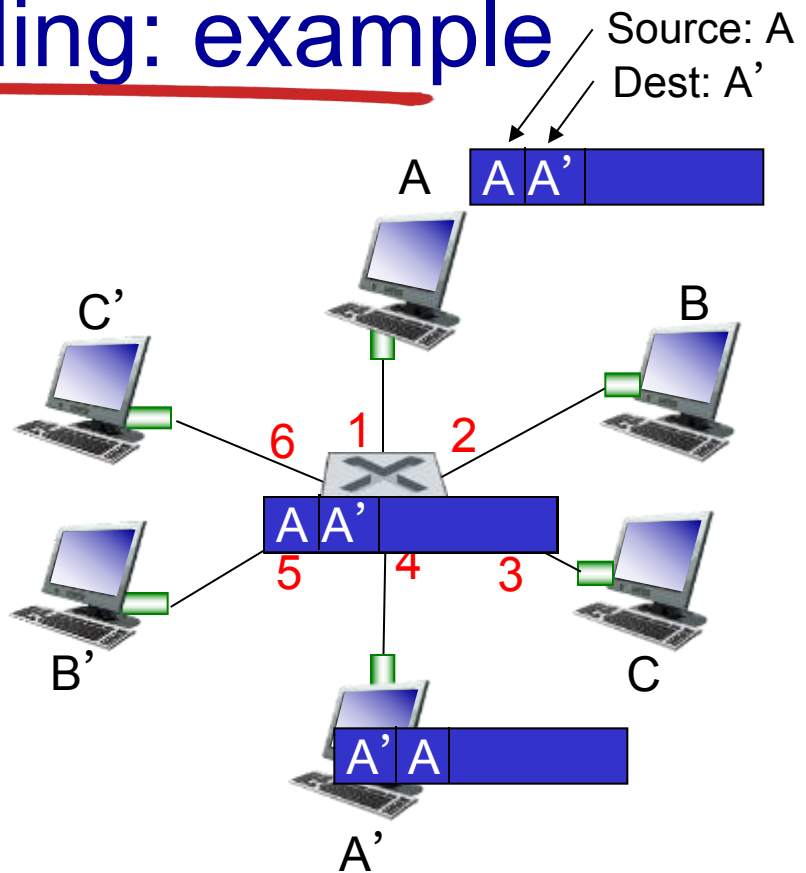


MAC addr	interface	TTL
A	1	60

*Switch table
(initially empty)*

Self-learning, forwarding: example

- frame destination, A', location unknown: *flood*
- destination A location known: *selectively send on just one link*



MAC addr	interface	TTL
A	1	60
A'	4	60

*switch table
(initially empty)*

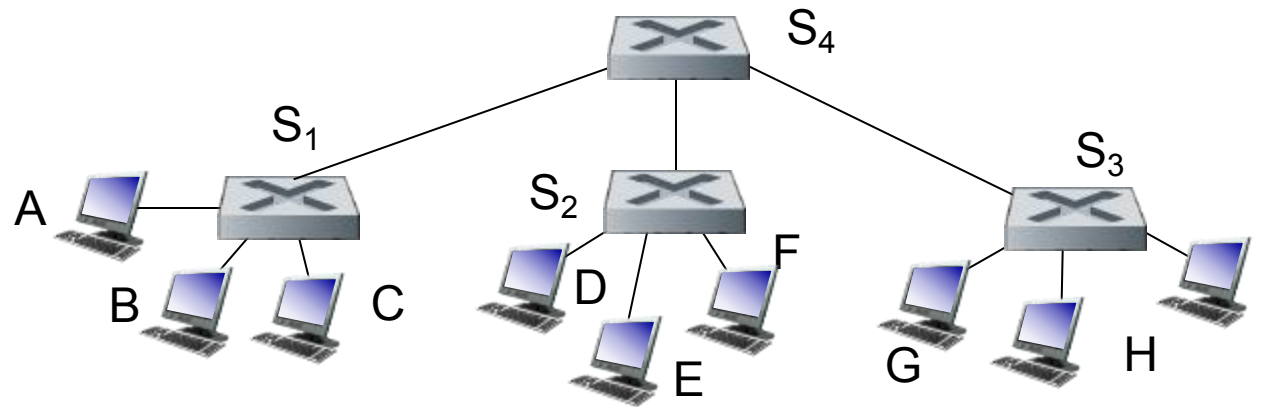
Self-learning, forwarding: example

Suppose a frame with destination address DDDD-DD-DD-DD-DD arrives at the switch **on interface x**.

- **no entry** in the table for DD-DD-DD-DD-DD-DD:
 - forwards copies of the frame to the output buffers preceding all interfaces except for interface x.
- an entry in the table associating DD-DD-DD-DD-DD-DD with **interface x**:
 - the switch performs the filtering function by discarding the frame.
- an entry in the table associating DD-DD-DD-DD-DD-DD with **interface y**:
 - frame needs to be forwarded to the LAN segment attached to interface y.

Interconnecting switches

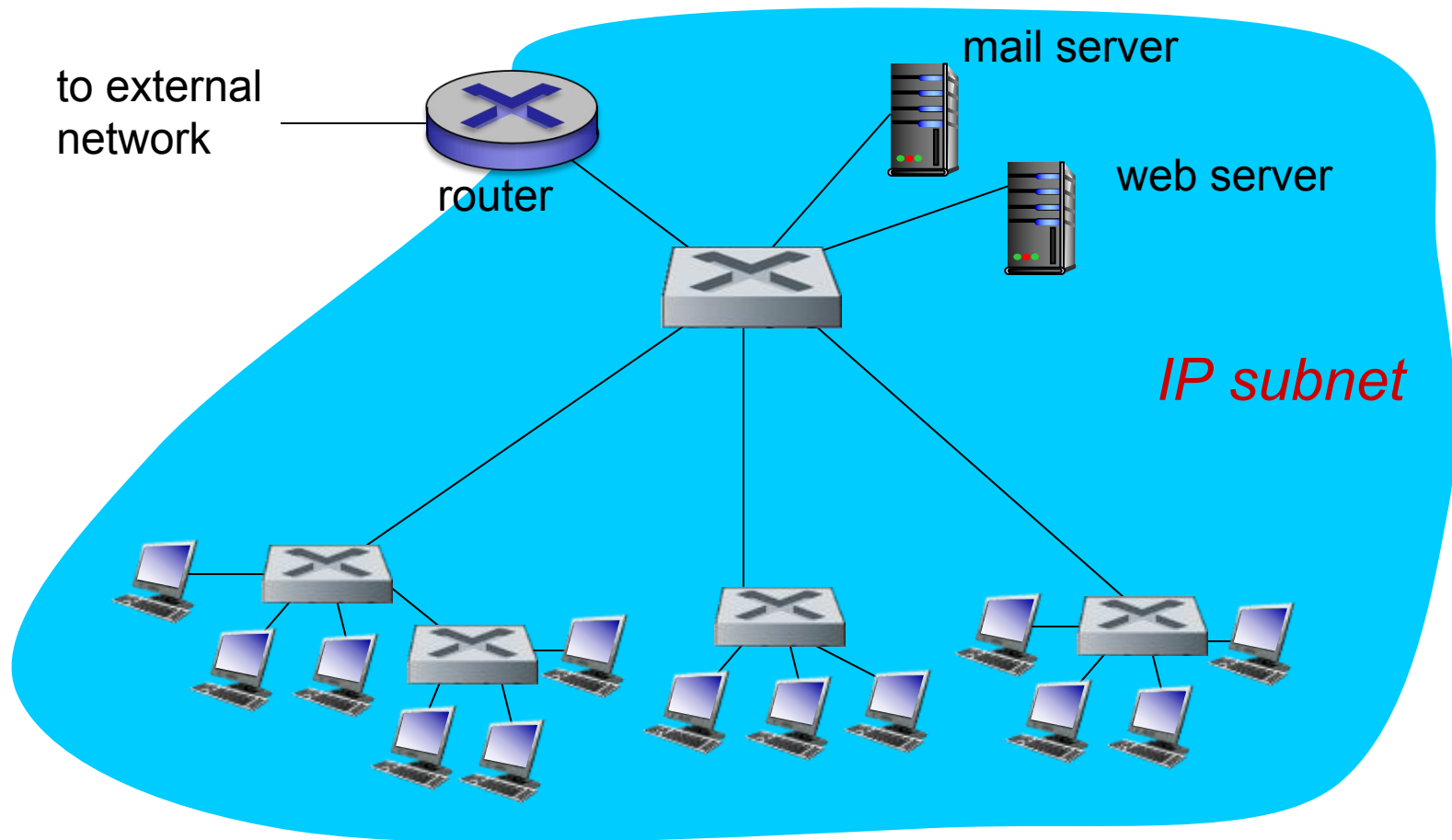
self-learning switches can be connected together:



Q: sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?

- A: self learning! (works exactly the same as in single-switch case!)

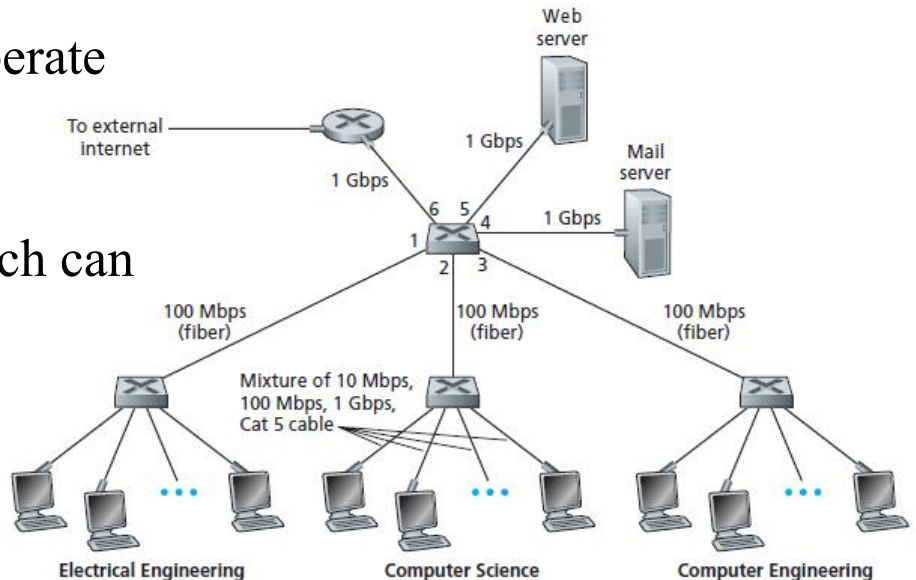
Institutional network



Properties of link-layer Switching

When compared with bus or hub:

- **Elimination of collisions**
 - Buffer frames; never transmit more than one frame on a segment at a time
- **Heterogeneous links**
 - Switch can isolate one link from another
 - Different links in the LAN can operate at different speeds and media
- **Management**
 - If one NIC malfunctions, the switch can detect it and disconnect it.



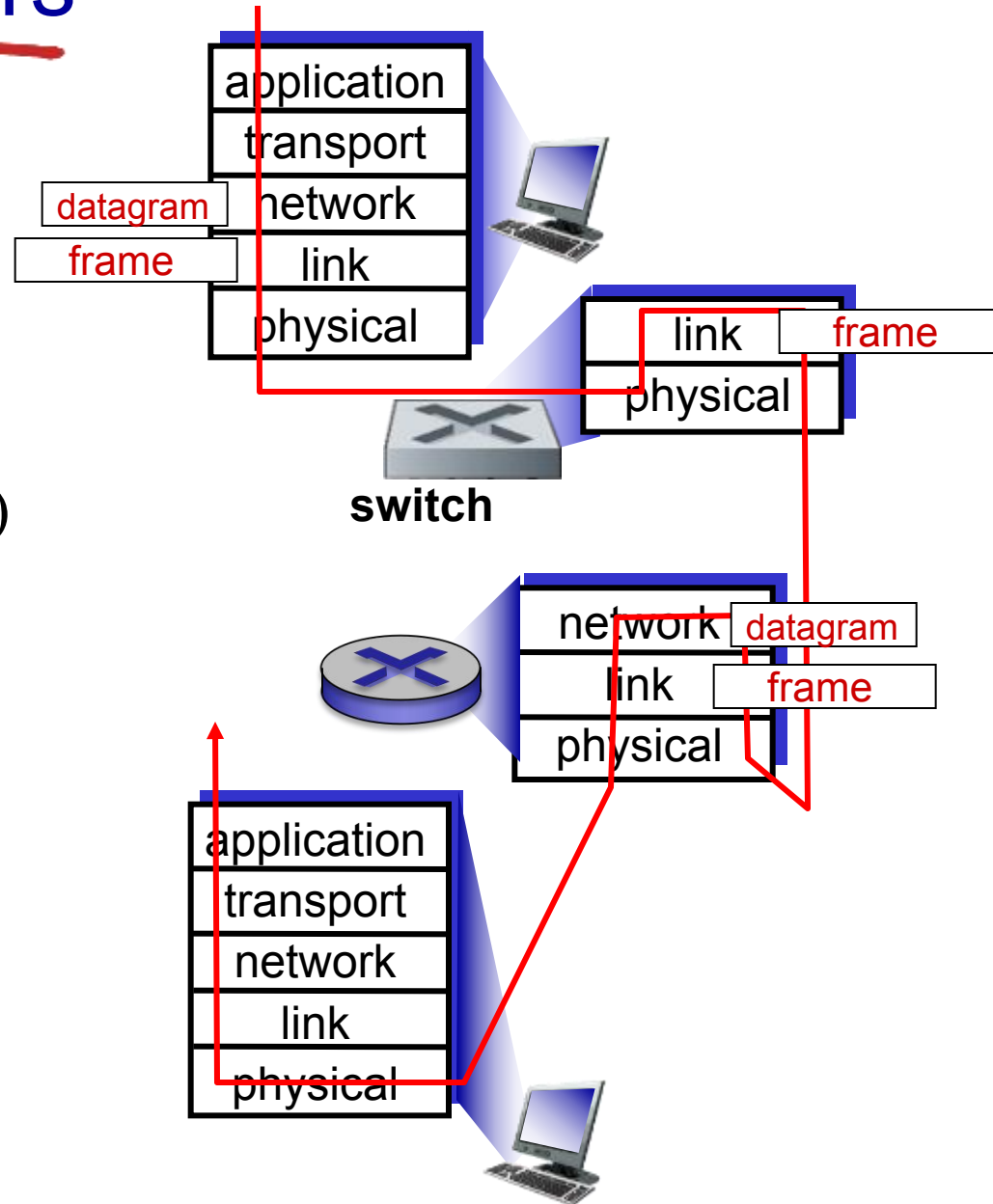
Switches vs. routers

both are store-and-forward:

- *routers*: network-layer devices (examine network-layer headers)
- *switches*: link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers*: compute tables using routing algorithms, IP addresses
- *switches*: learn forwarding table using flooding, learning, MAC addresses



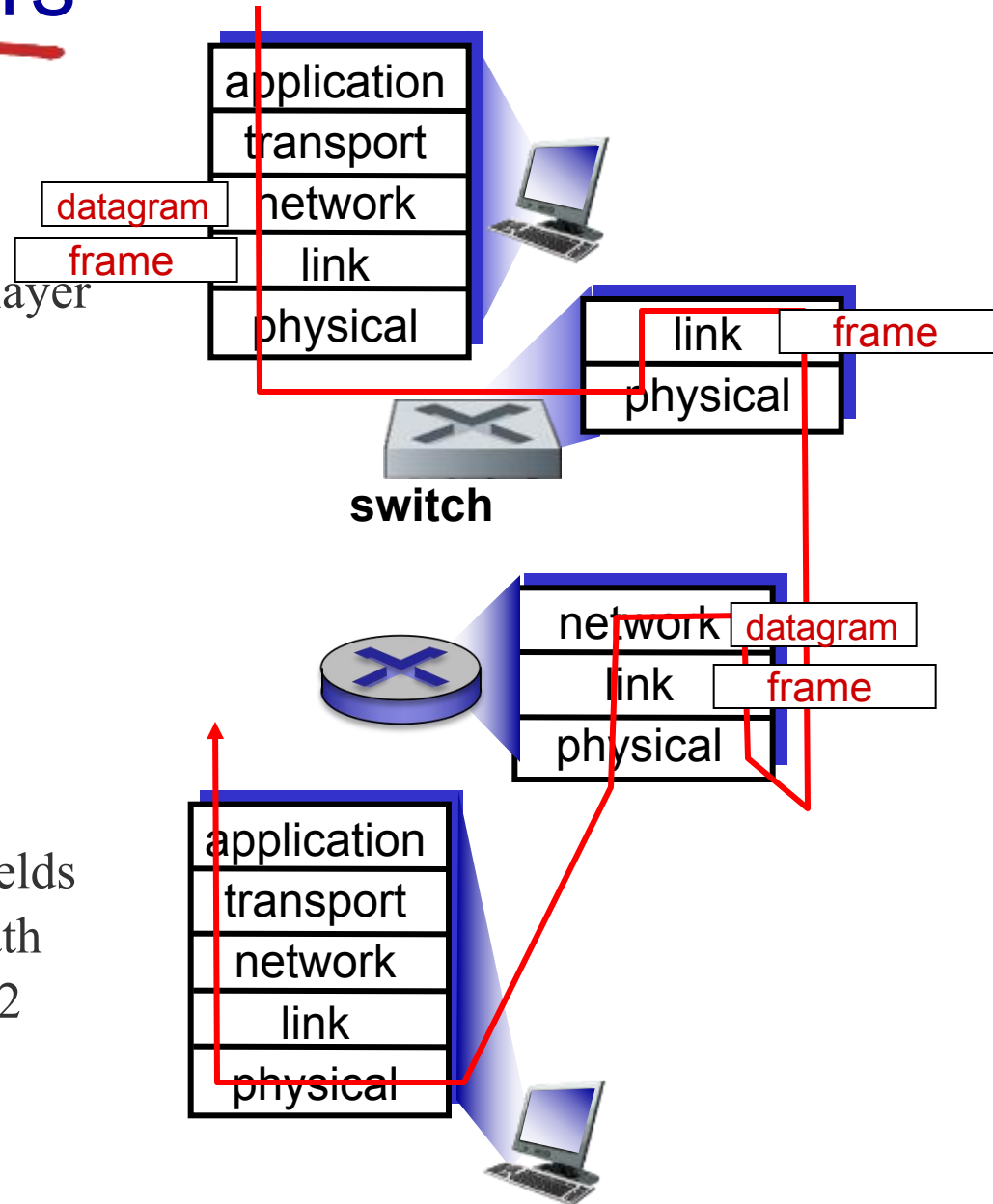
Switches vs. routers

Switches:

- plug-and-play
- Process frames only up through layer 2, relatively high filtering and forwarding rates
- a spanning tree
- a large switched network would require large ARP
- broadcast storms

Routers

- not plug-and-play
- process up through the layer-3 fields
- rich topology; choose the best path
- firewall protection against layer-2 broadcast storms.



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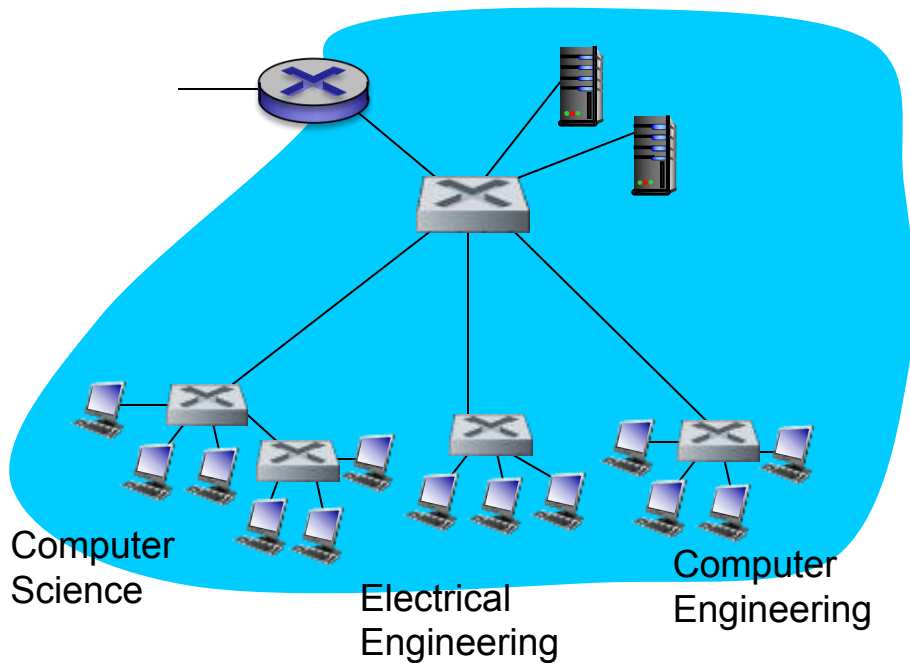
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VLANs: motivation



consider:

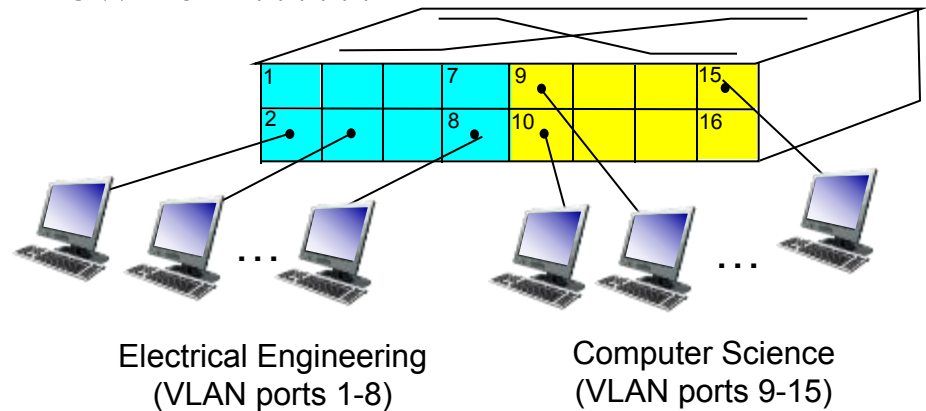
- CS user moves office to EE, but wants to connect to CS switch
- Inefficient use of switches
- single broadcast domain:
 - all layer-2 broadcast traffic (ARP, DHCP, unknown location of destination MAC address) must cross entire LAN
 - security/privacy, efficiency issues

VLANs

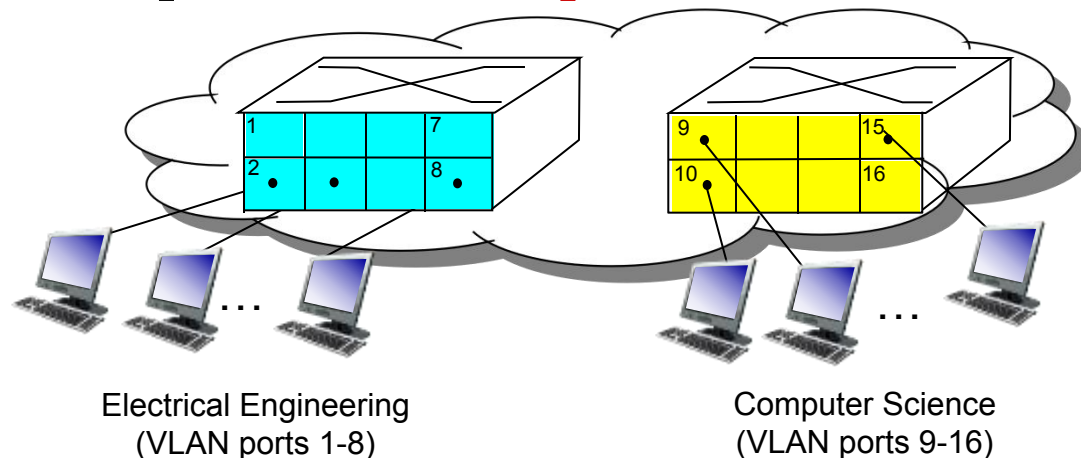
Virtual Local Area Network

define multiple *virtual* LANS over single physical LAN infrastructure.

port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch

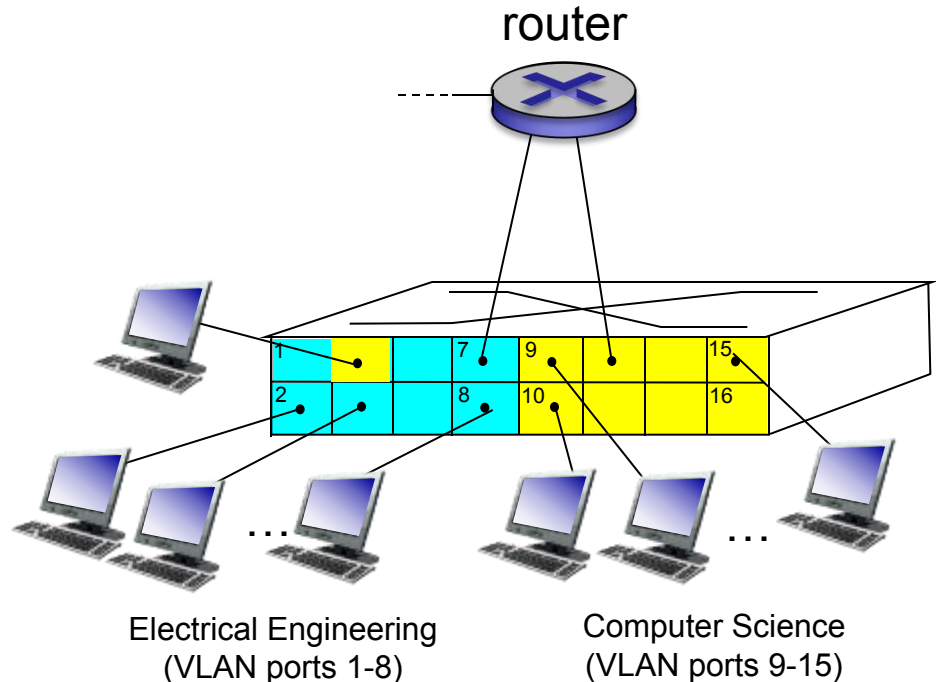


... operates as **multiple** virtual switches

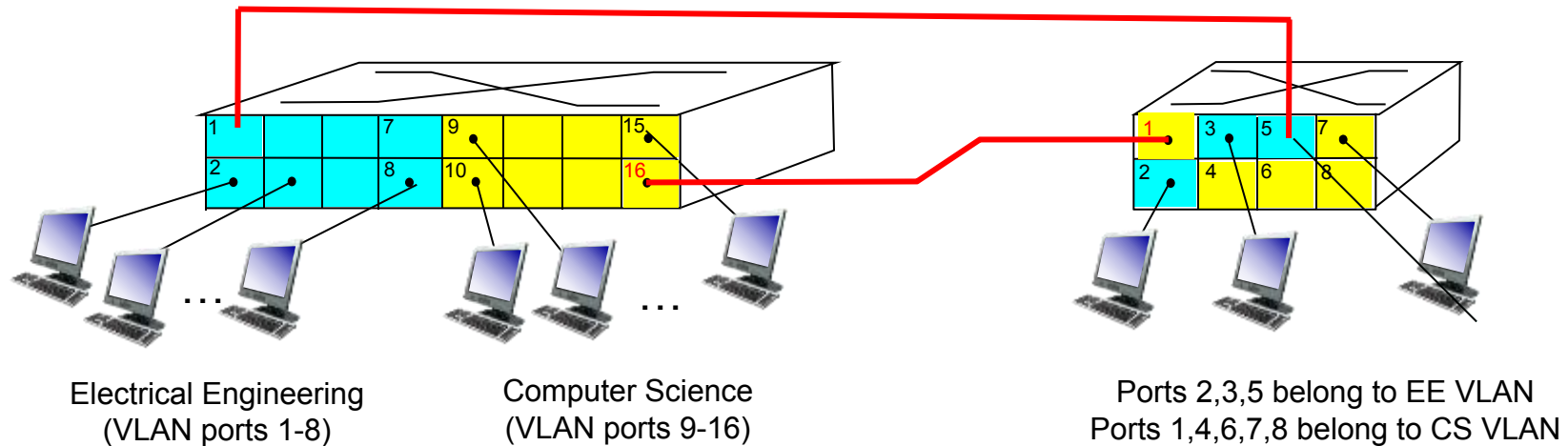


Port-based VLAN

- **traffic isolation:** frames to/from ports 1-8 can *only* reach ports 1-8
 - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- **dynamic membership:** ports can be dynamically assigned among VLANs
- **forwarding between VLANs:** done via routing (just as with separate switches)
 - in practice vendors sell combined switches plus routers

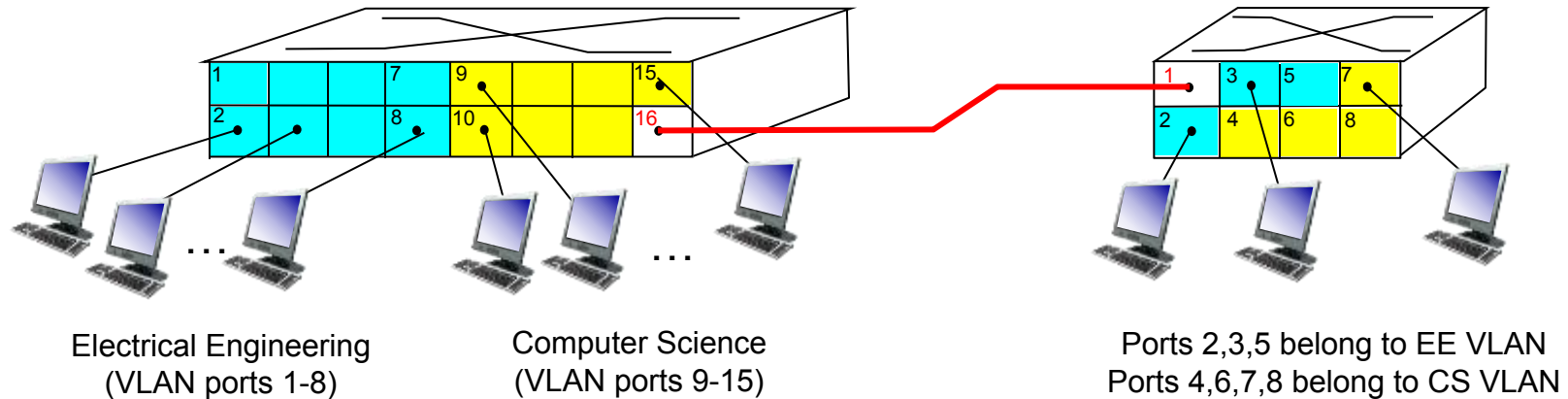


VLANs spanning multiple switches



- *If some of the CS and EE faculties are in another building, how to connect two switches together as two VLANs?*
 - *Two links connect both CS VLAN and EE VLAN.*

VLANs spanning multiple switches



- **trunk port:** carries frames between VLANs defined over multiple physical switches
 - frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry VLAN ID info)
 - Extended Ethernet frame format: 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

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Data center networks

- 10's to 100's of thousands of hosts, often closely coupled, in close proximity:
 - e-business (e.g. Amazon)
 - content-servers (e.g., YouTube, Akamai, Apple, Microsoft)
 - search engines, data mining (e.g., Google)
- challenges:
 - multiple applications, each serving massive numbers of clients
 - managing/balancing load, avoiding processing, networking, data bottlenecks

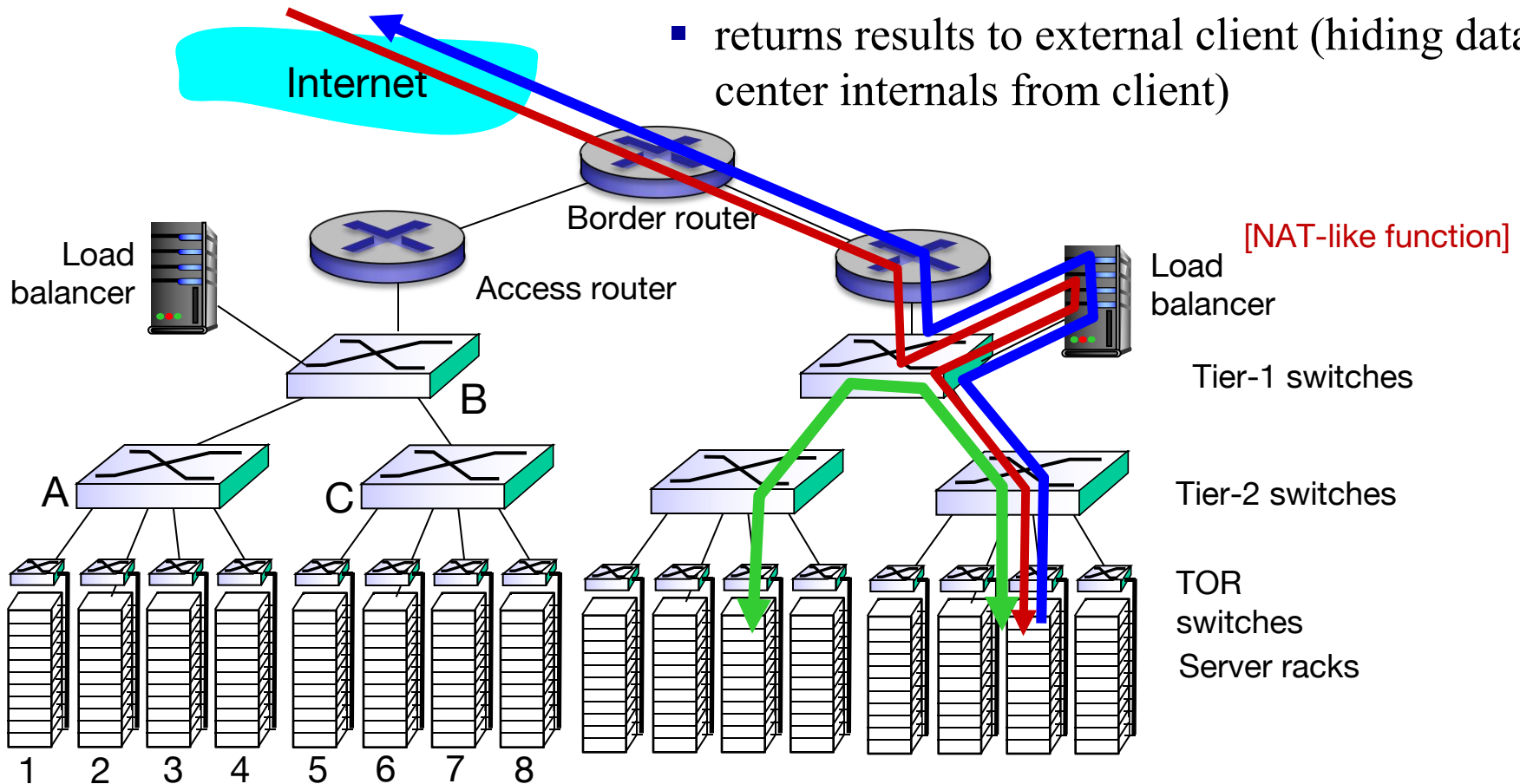


Inside a 40-ft Microsoft container,
Chicago data center

Data center networks

load balancer: application-layer routing

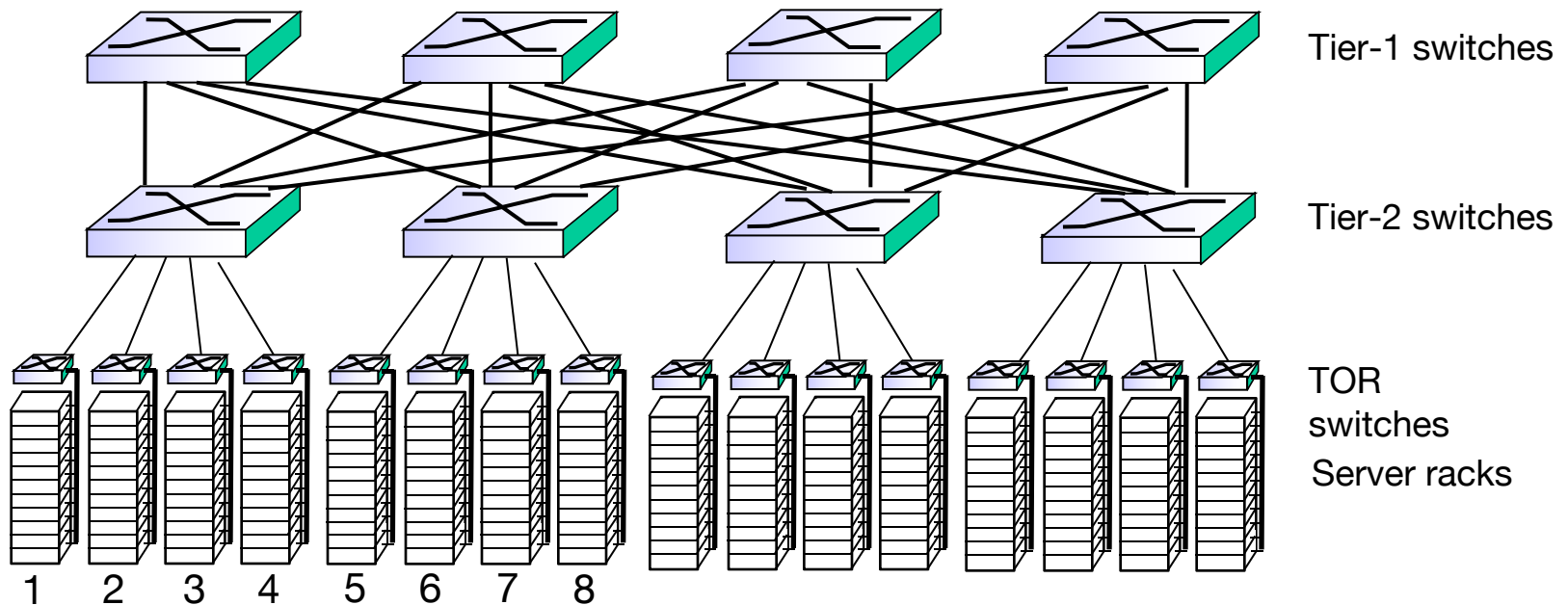
- receives external client requests
- directs workload within data center
- returns results to external client (hiding data center internals from client)



(Hosts are stacked in racks)

Data center networks

- rich interconnection among switches, racks:
 - increased throughput between racks (multiple routing paths possible)
 - increased reliability via redundancy



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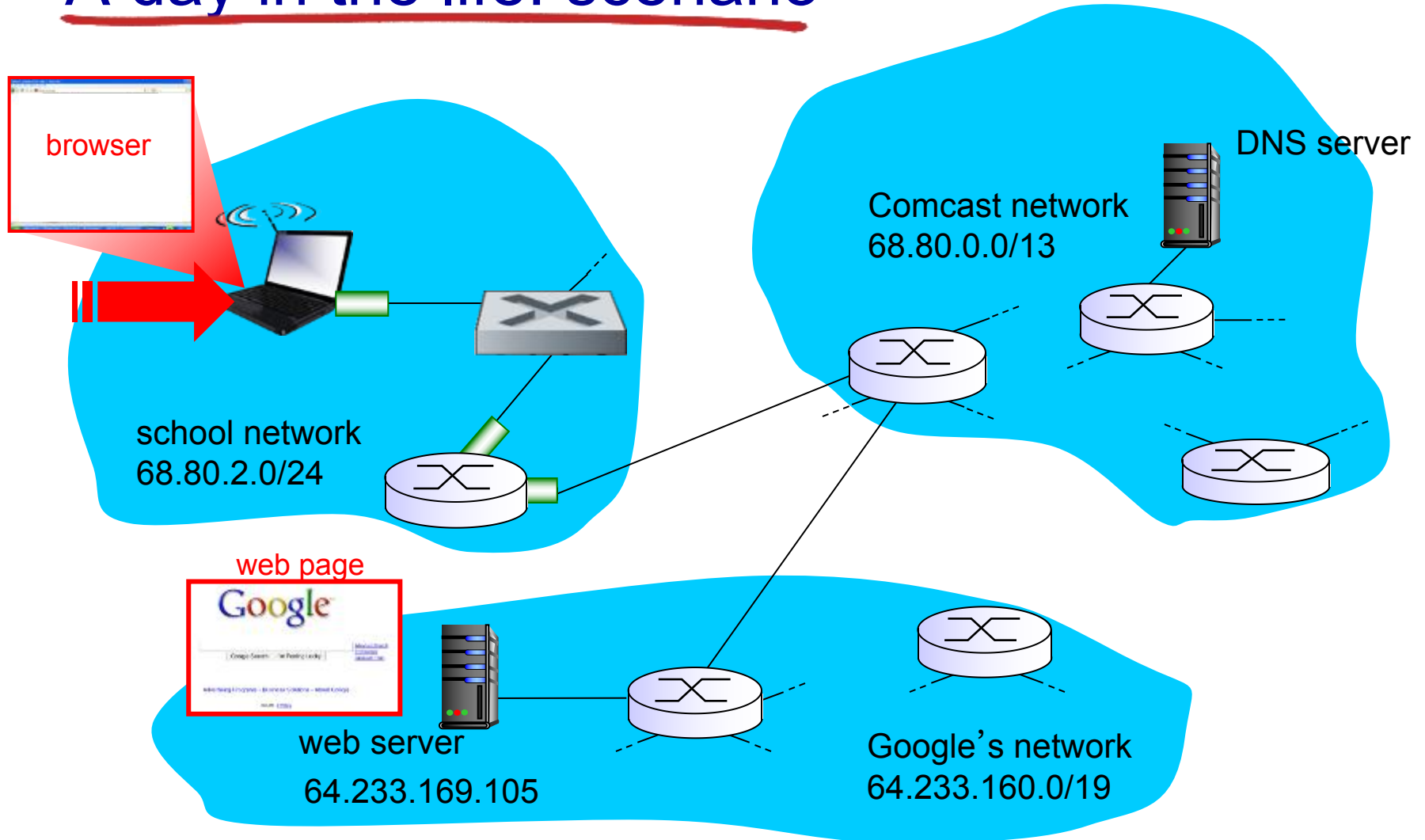
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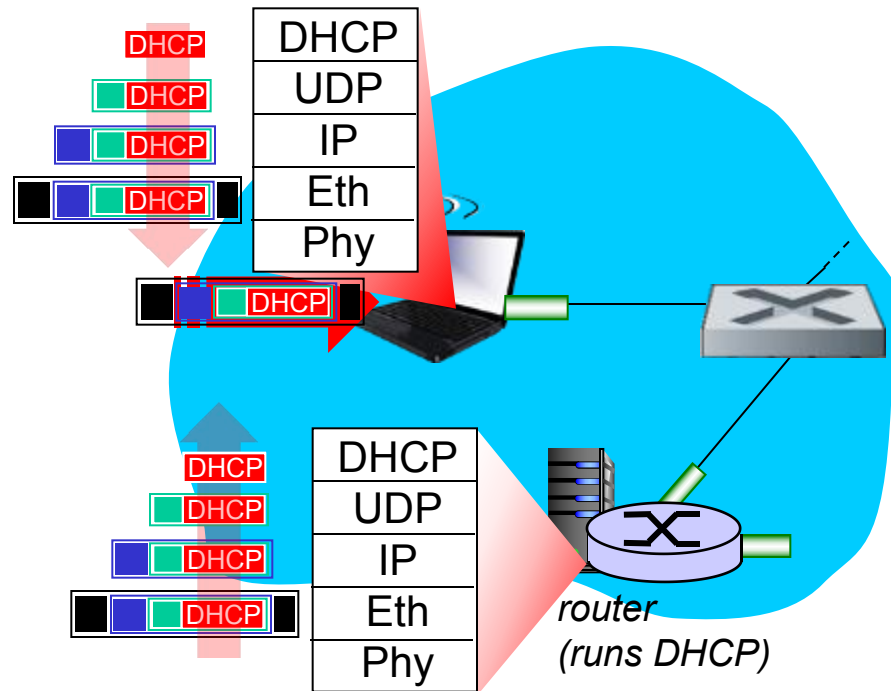
Synthesis: a day in the life of a web request

- journey down protocol stack complete!
 - application, transport, network, link
- putting-it-all-together: synthesis!
 - *goal*: identify, review, understand protocols (at all layers) involved in seemingly simple scenario: requesting www page
 - *scenario*: student attaches laptop to campus network, requests/receives www.google.com

A day in the life: scenario

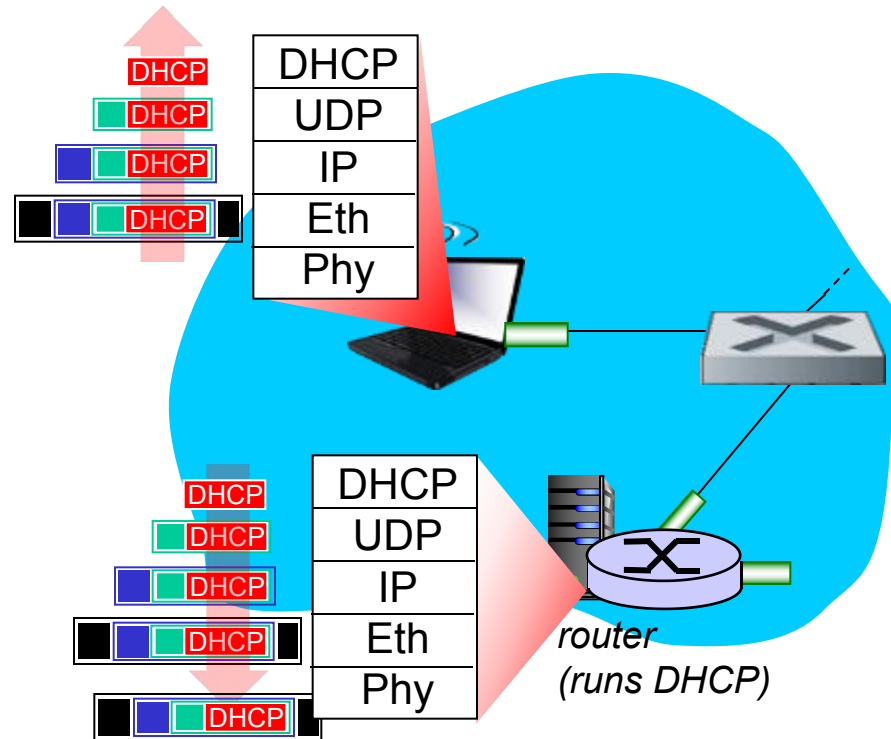


A day in the life... connecting to the Internet



- connecting laptop needs to get its own IP address, addr of first-hop router, addr of DNS server: use *DHCP*
- DHCP request **encapsulated** in **UDP**, encapsulated in **IP**, encapsulated in **802.3** Ethernet
- Ethernet frame **broadcast** (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running **DHCP** server
- Ethernet **demuxed** to IP demuxed, UDP demuxed to DHCP

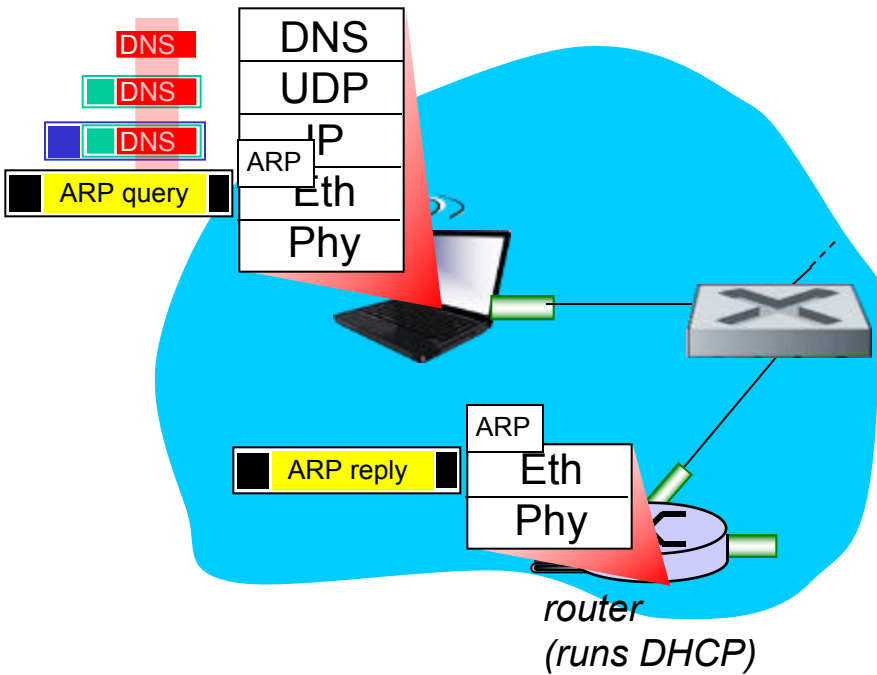
A day in the life... connecting to the Internet



- DHCP server formulates *DHCP ACK* containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulation at DHCP server, frame forwarded (*switch learning*) through LAN, demultiplexing at client
- DHCP client receives DHCP ACK reply

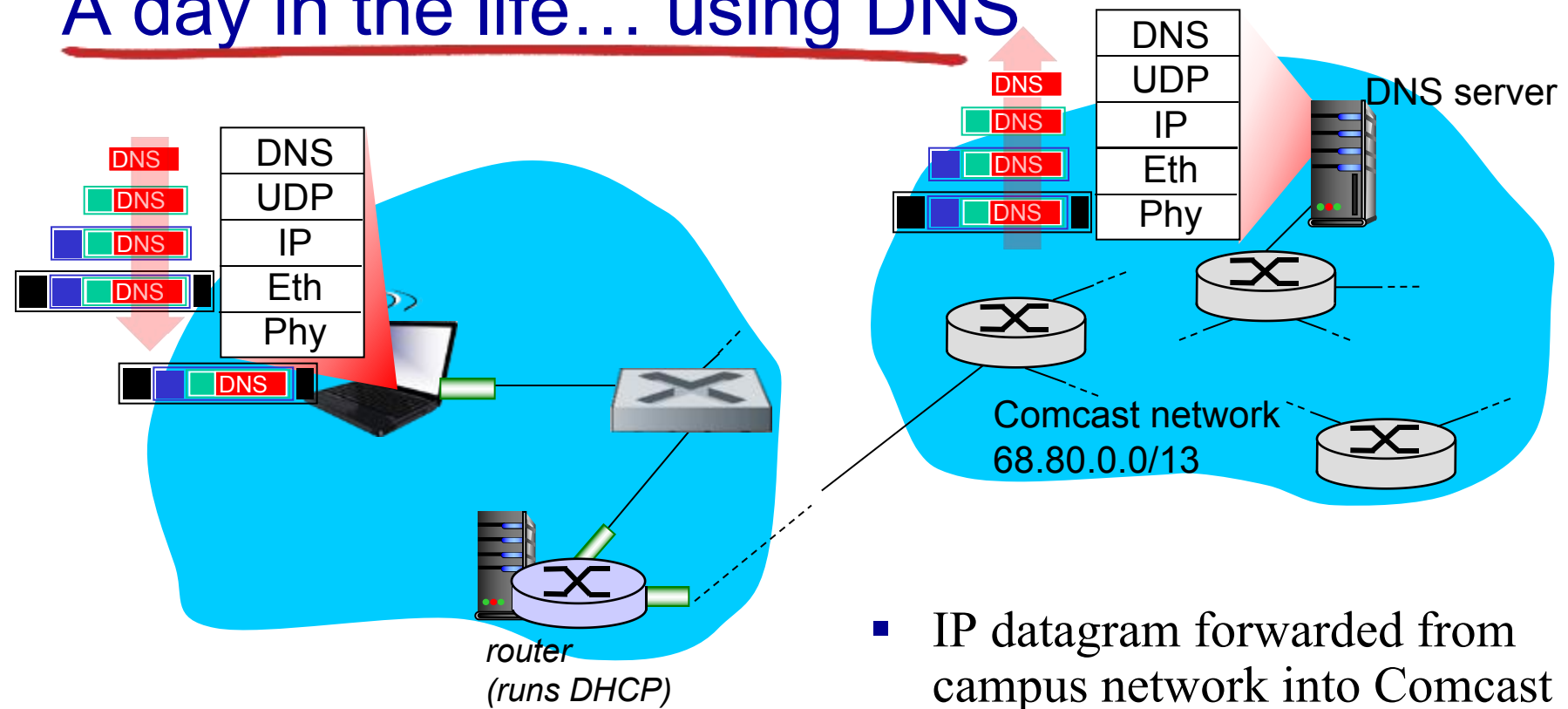
Client now has IP address, knows name & addr of DNS server, IP address of its first-hop router

A day in the life... ARP (before DNS, before HTTP)



- before sending *HTTP* request, need IP address of `www.google.com`:
DNS
- DNS query created, encapsulated in UDP, encapsulated in IP, encapsulated in Eth. To send frame to router, need MAC address of router interface: *ARP*
- *ARP query* broadcast, received by router, which replies with *ARP reply* giving MAC address of router interface
- client now knows MAC address of first hop router, so can now send frame containing DNS query

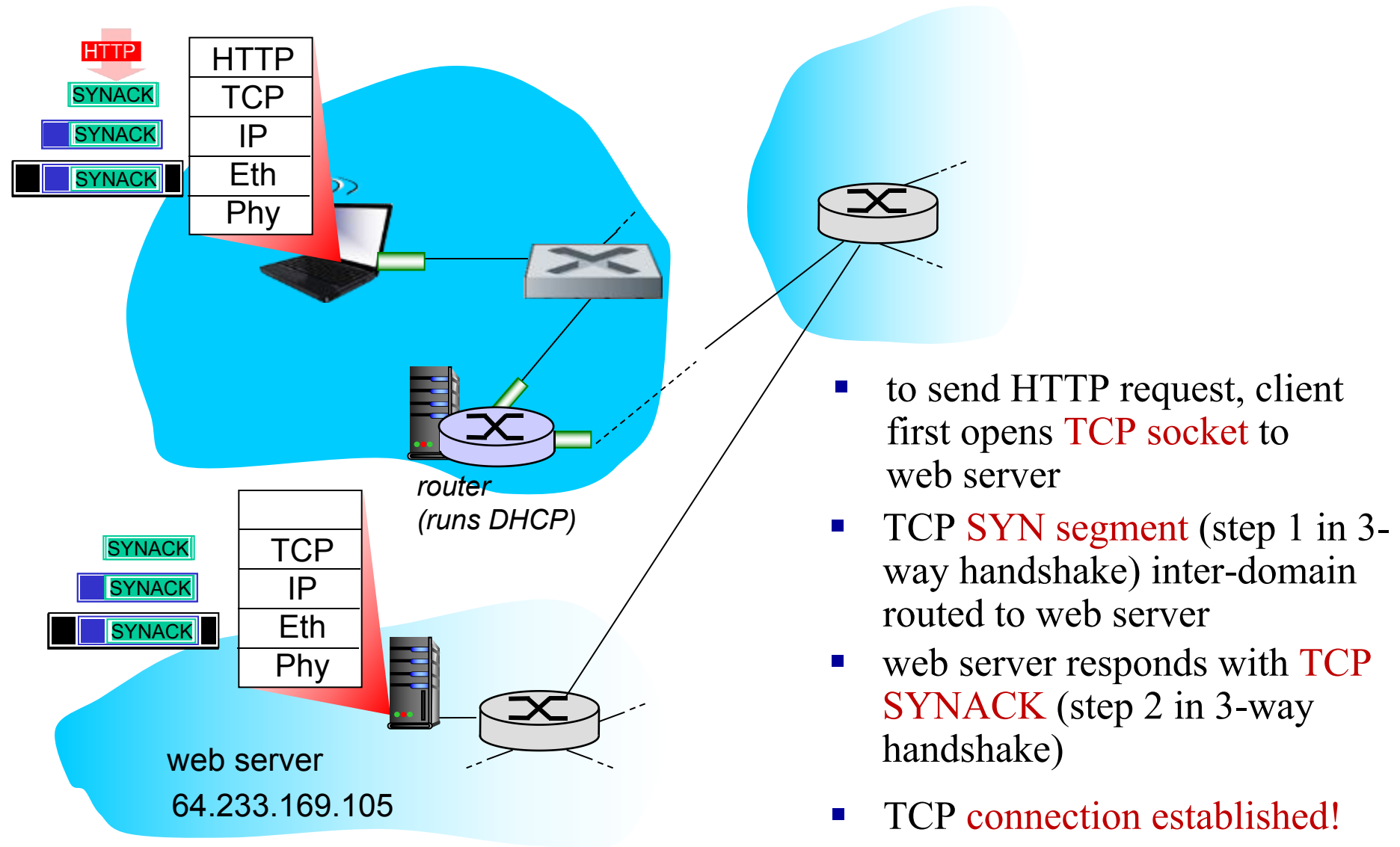
A day in the life... using DNS



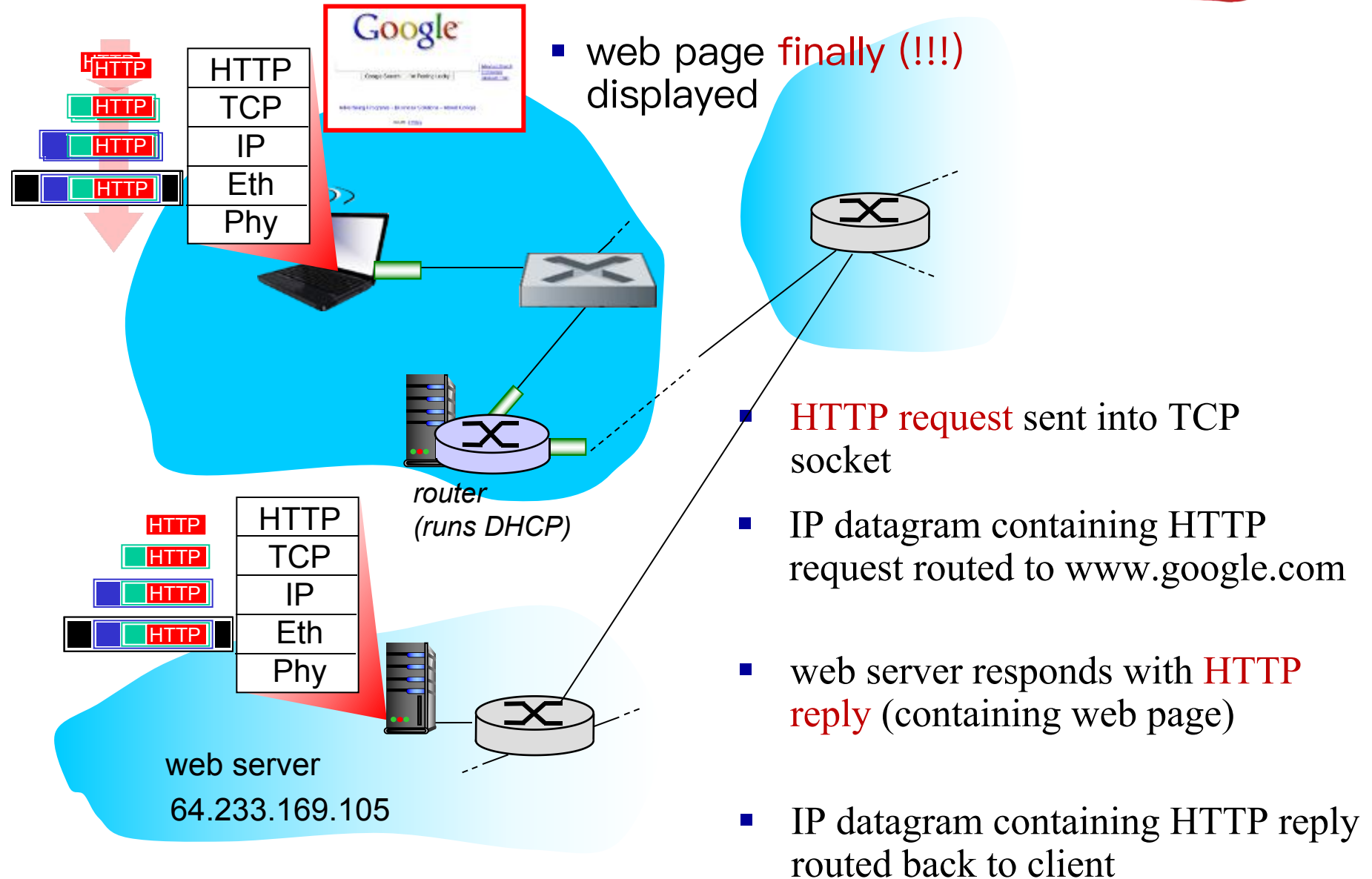
- IP datagram containing DNS query forwarded via LAN switch from client to 1st hop router

- IP datagram forwarded from campus network into Comcast network, routed (tables created by **RIP**, **OSPF**, **IS-IS** and/or **BGP** routing protocols) to DNS server
- demuxed to DNS server
- DNS server replies to client with IP address of www.google.com

A day in the life...TCP connection carrying HTTP



A day in the life... HTTP request/reply



Ch. 7: Wireless and Mobile Networks

Background:

- Number of wireless (mobile) phone subscribers now exceeds number of wired phone subscribers (5-to-1)!
- Number of wireless Internet-connected devices equals number of wireline Internet-connected devices
 - laptops, Internet-enabled phones promise anytime untethered Internet access
- two important (but different) challenges
 - *wireless*: communication over wireless link
 - *mobility*: handling the mobile user who changes point of attachment to network

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 - *mobility*: handling the mobile user who changes point of attachment to network

Chapter 7 outline

7.1 Introduction

Wireless

7.2 Wireless links, characteristics

- CDMA

7.3 IEEE 802.11 wireless LANs (“Wi-Fi”)

7.4 Cellular Internet Access

- architecture
- standards (e.g., 3G, LTE)

Mobility

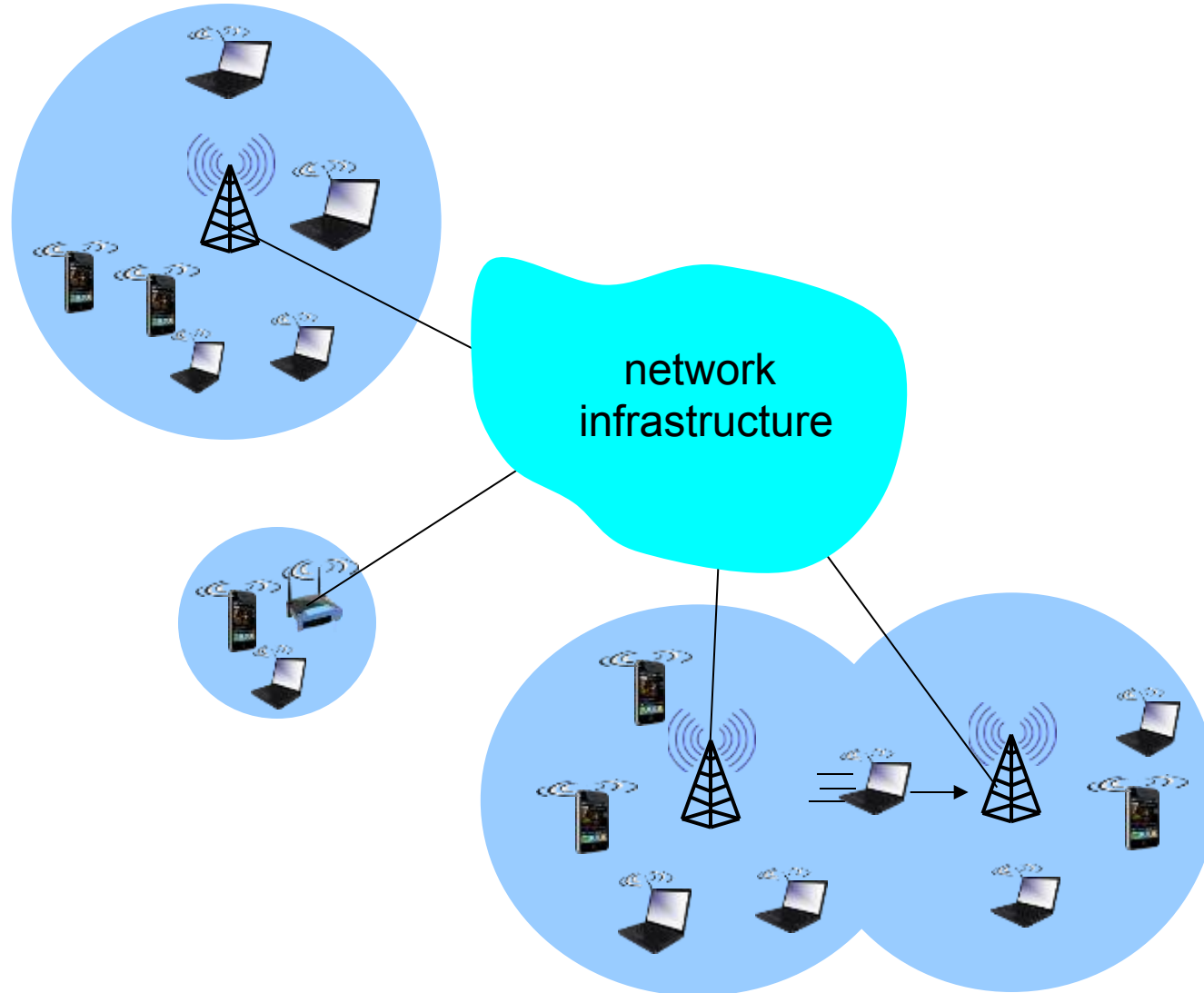
7.5 Principles: addressing and routing to mobile users

7.6 Mobile IP

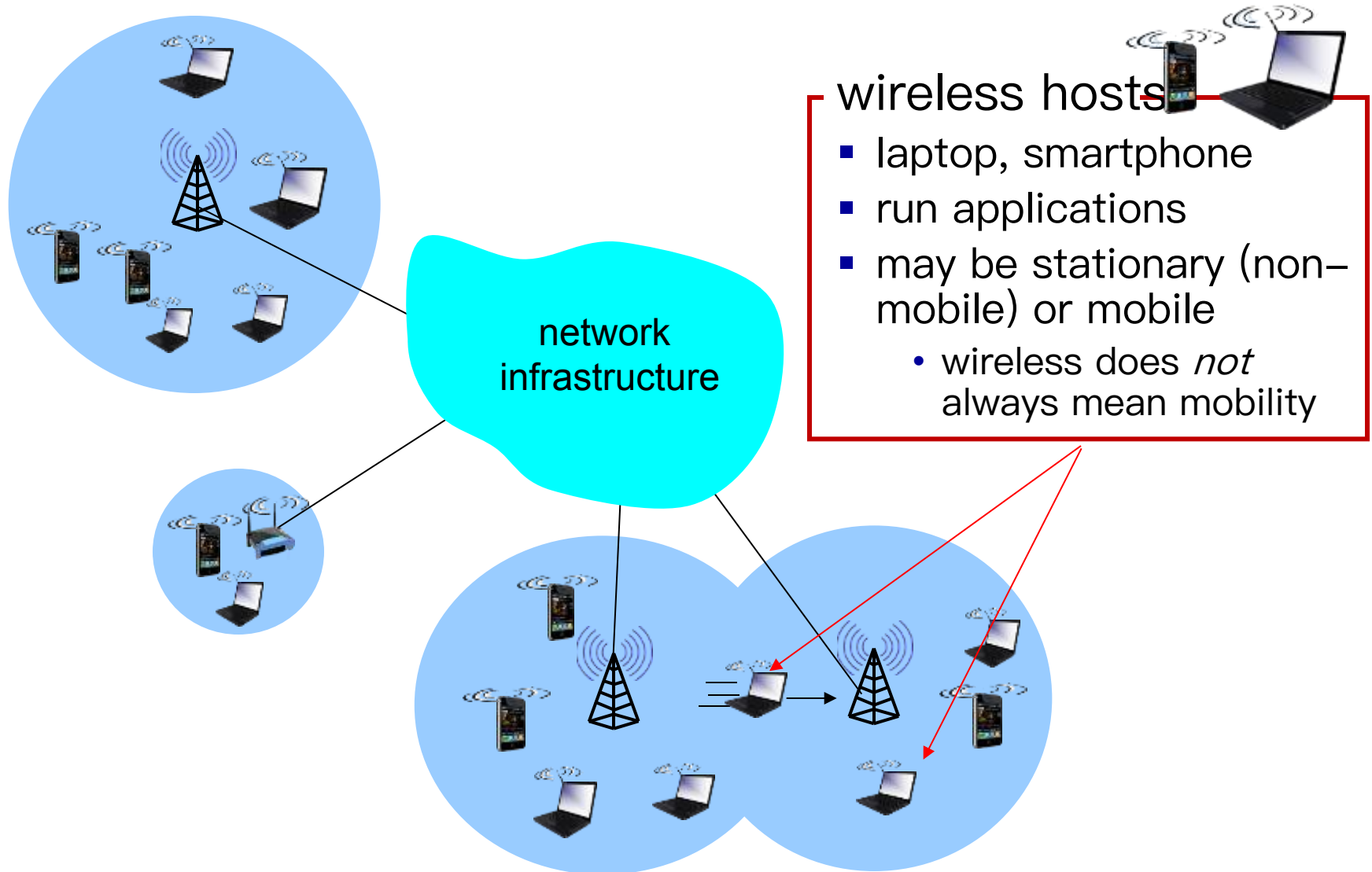
7.7 Handling mobility in cellular networks

7.8 Mobility and higher-layer protocols

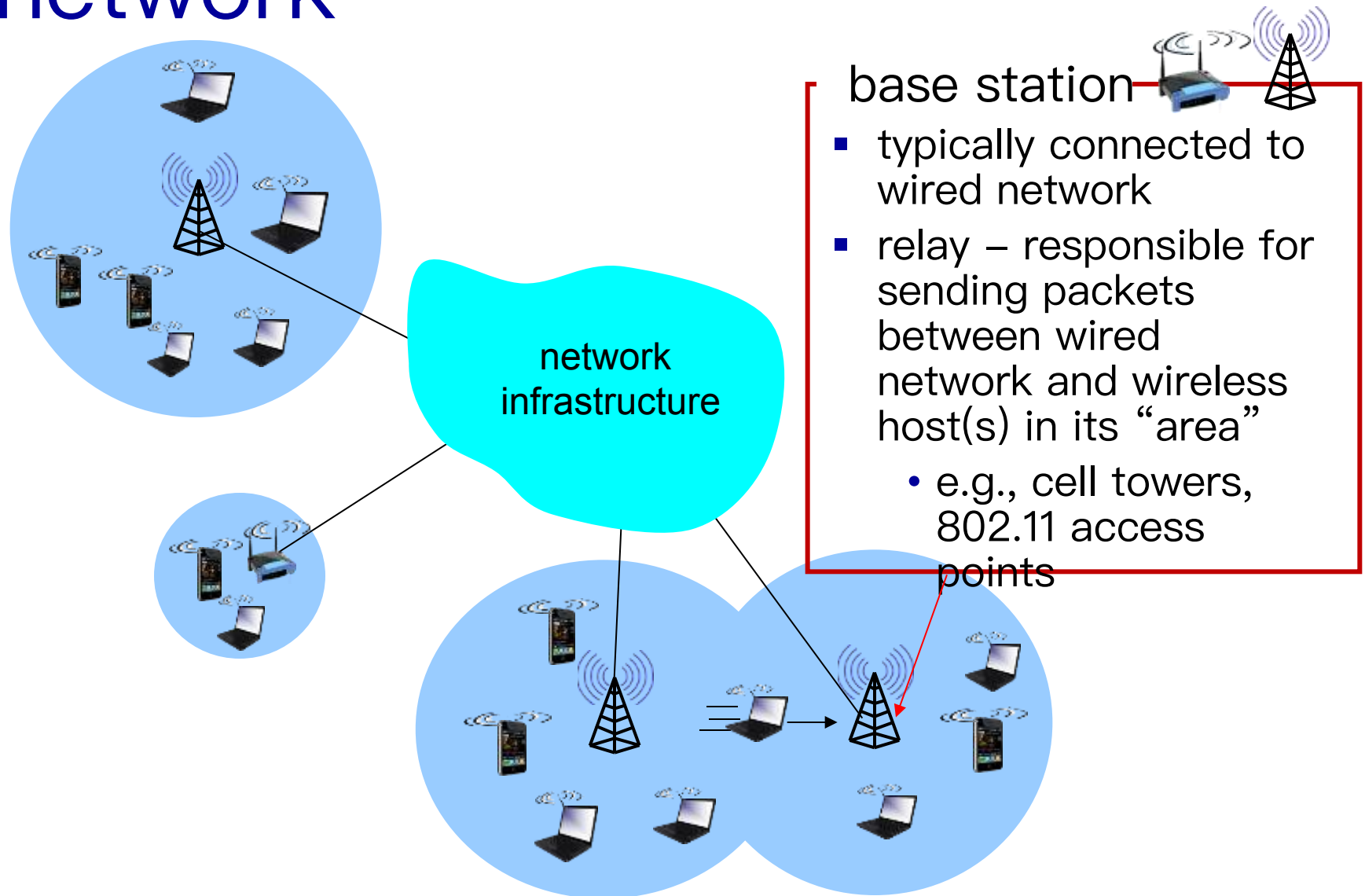
Elements of a wireless network



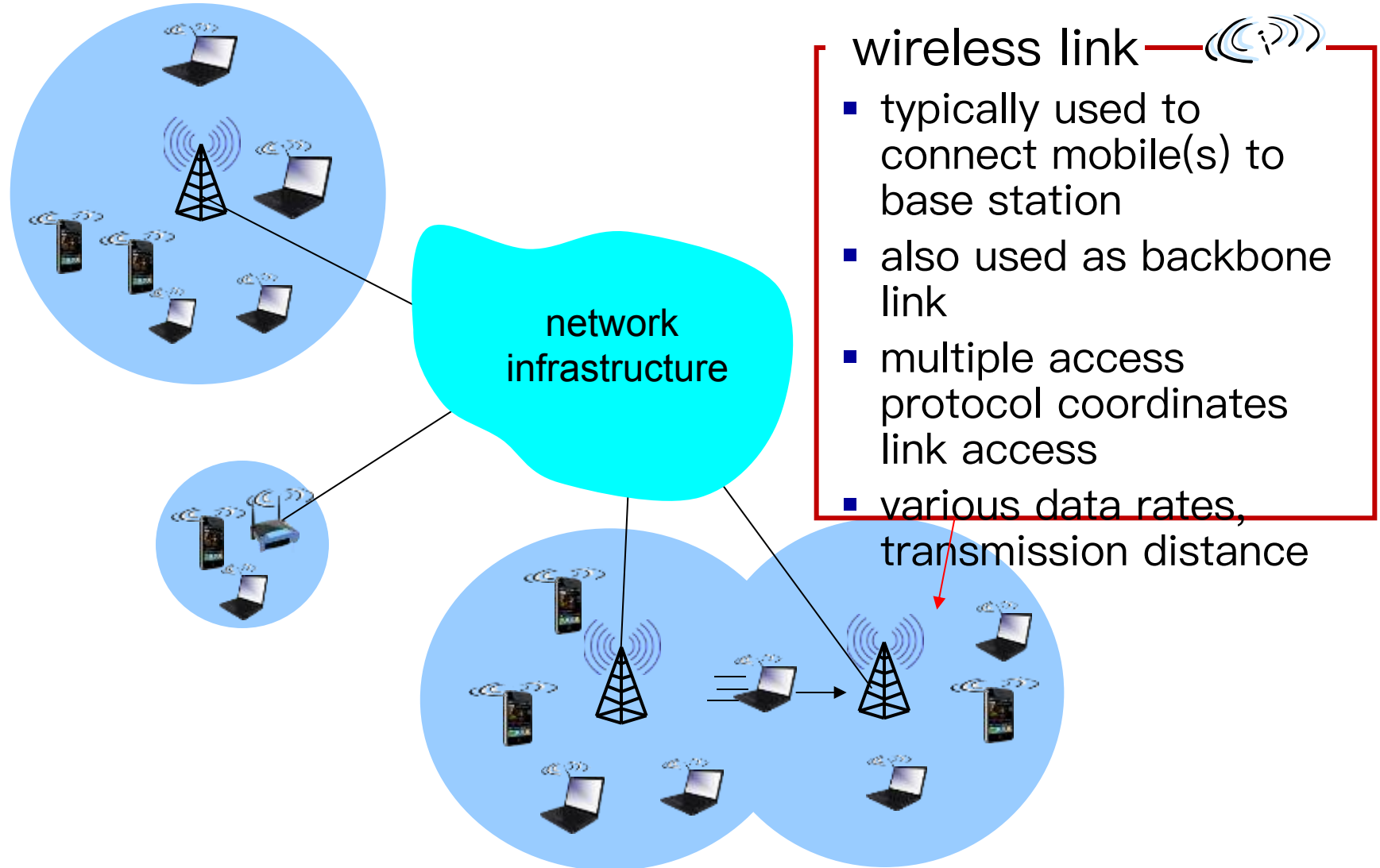
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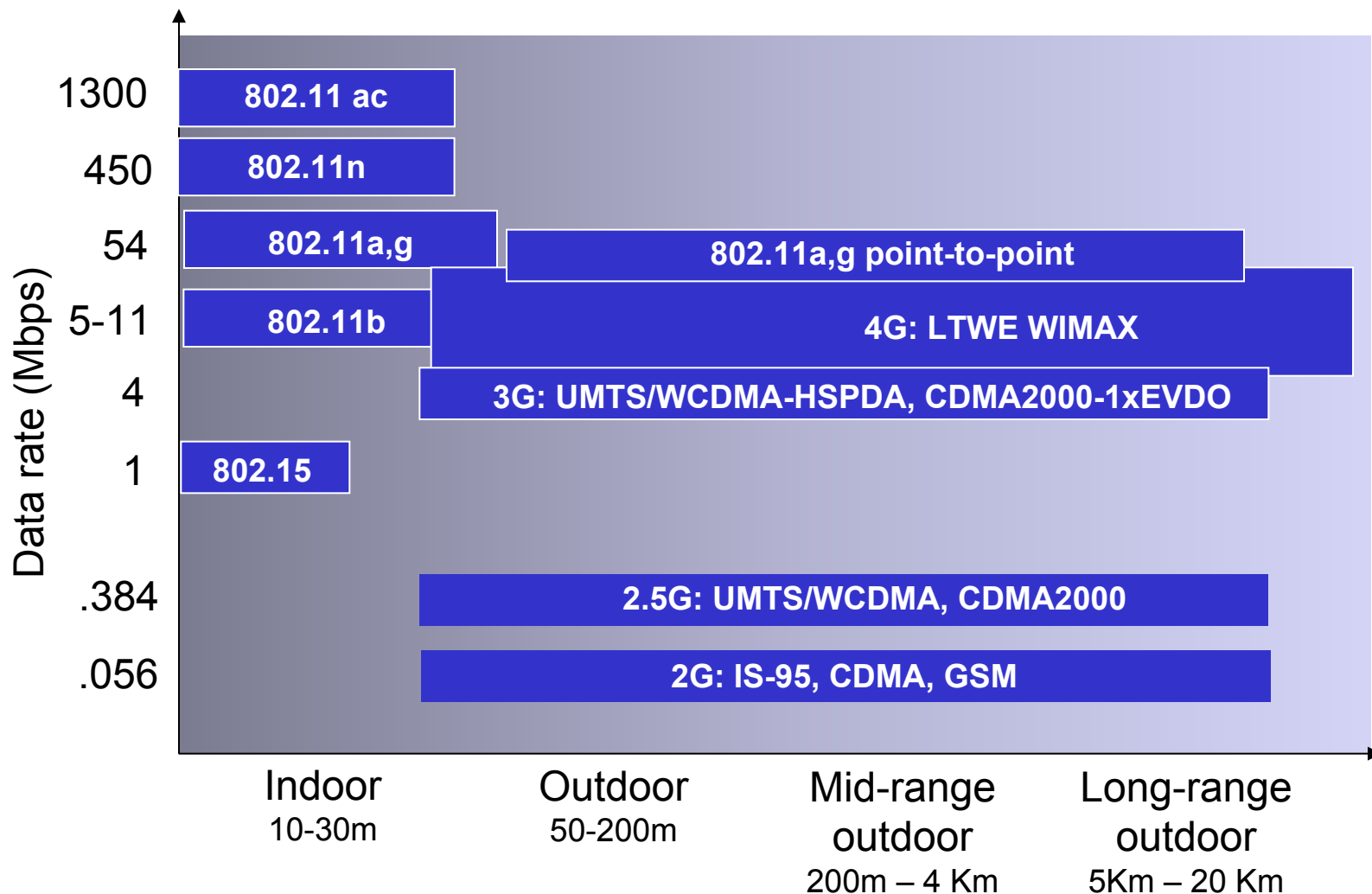
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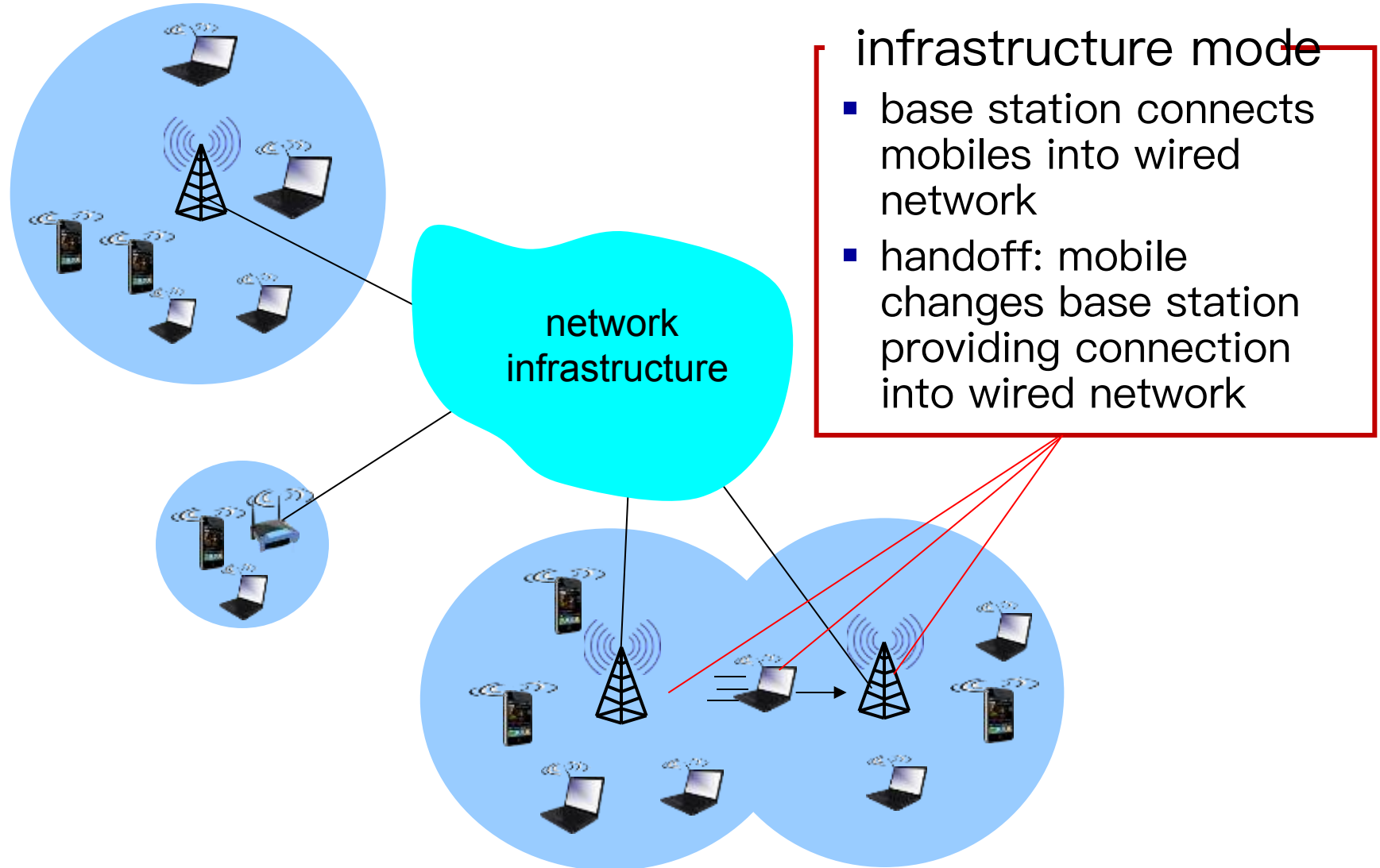
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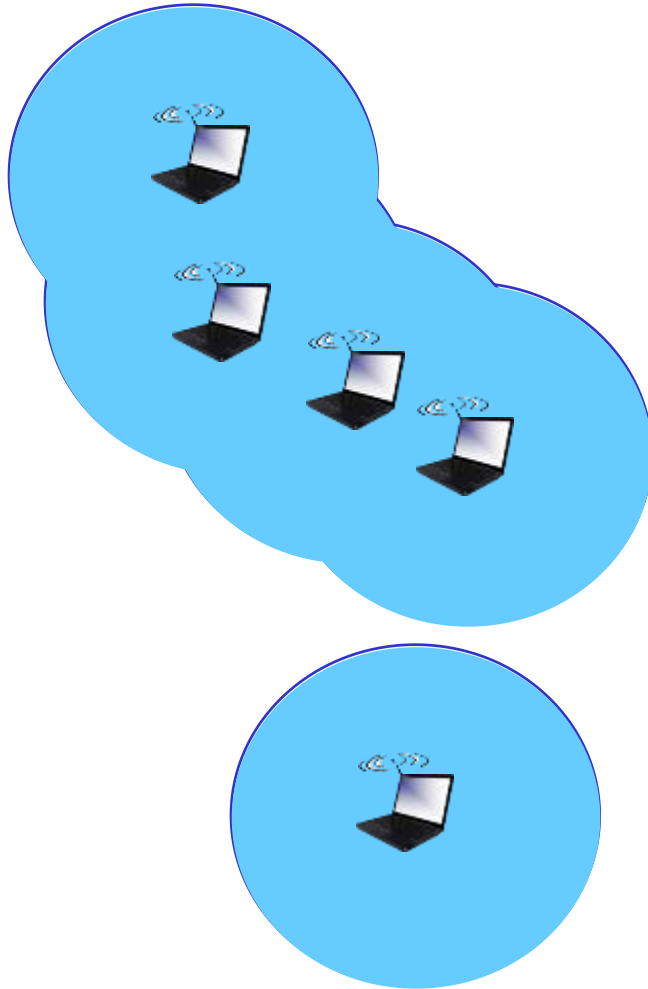
Characteristics of selected wireless links



Elements of a wireless network



Elements of a wireless network



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route among themselves

Wireless network taxonomy

	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, WiMAX, cellular) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet: <i>mesh net</i>
no infrastructure	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET

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Wireless Link Characteristics (1)

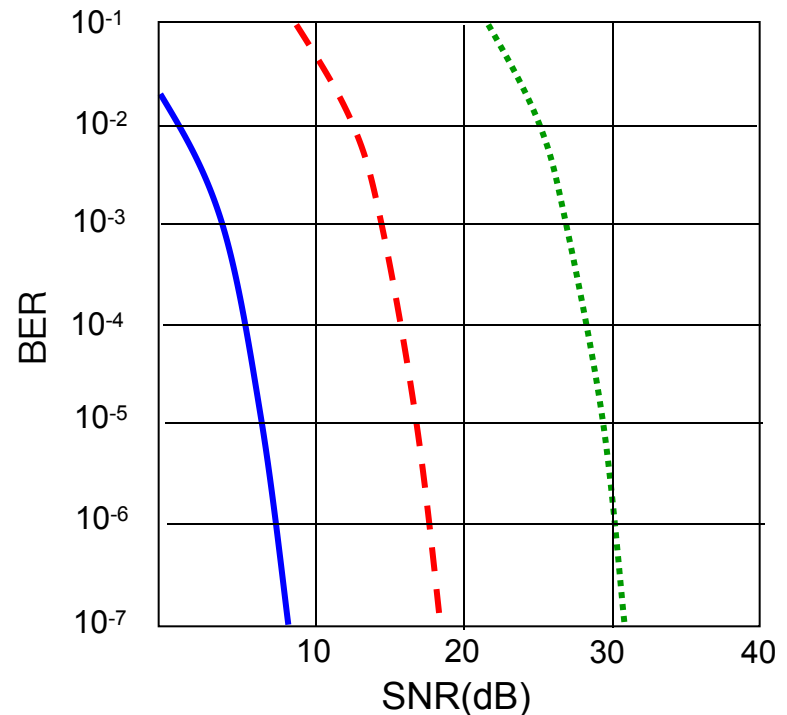
important differences from wired link

- *decreased signal strength*: radio signal attenuates as it propagates through matter (path loss)
- *interference from other sources*: standardized wireless network frequencies (e.g., 2.4 GHz) shared by other devices (e.g., phone); devices (motors) interfere as well
- *multipath propagation*: radio signal reflects off objects ground, arriving at destination at slightly different times

.... make communication across (even a point to point) wireless link much more “difficult”

Wireless Link Characteristics (2)

- SNR: signal-to-noise ratio
 - larger SNR – easier to extract signal from noise (a “good thing”)
- *SNR versus BER tradeoffs*
 - *given physical layer*: increase power $\rightarrow$ increase SNR $\rightarrow$ decrease BER
 - *given SNR*: choose physical layer that meets BER requirement, giving highest throughput
 - SNR may change with mobility: dynamically adapt physical layer (modulation technique, rate)



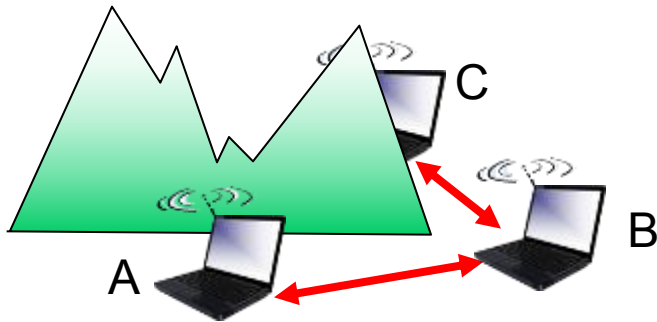
..... QAM256 (8 Mbps)

- - - QAM16 (4 Mbps)

— BPSK (1 Mbps)

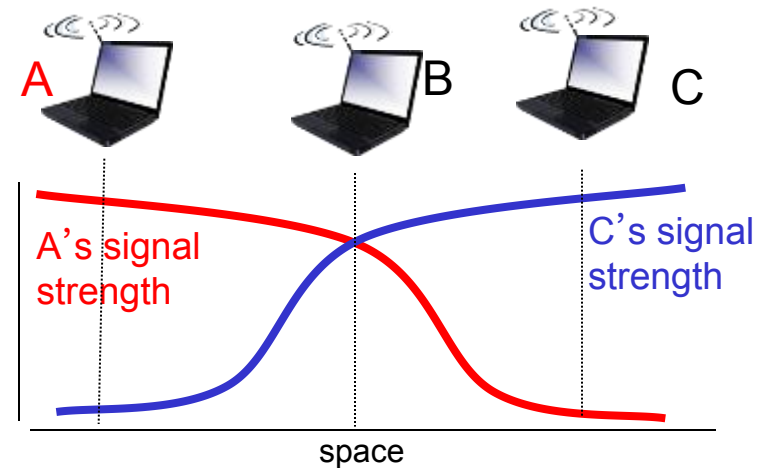
Wireless network characteristics

Multiple wireless senders and receivers create additional problems (beyond multiple access):



Hidden terminal problem

- B, A hear each other
- B, C hear each other
- A, C can not hear each other means A, C unaware of their interference at B



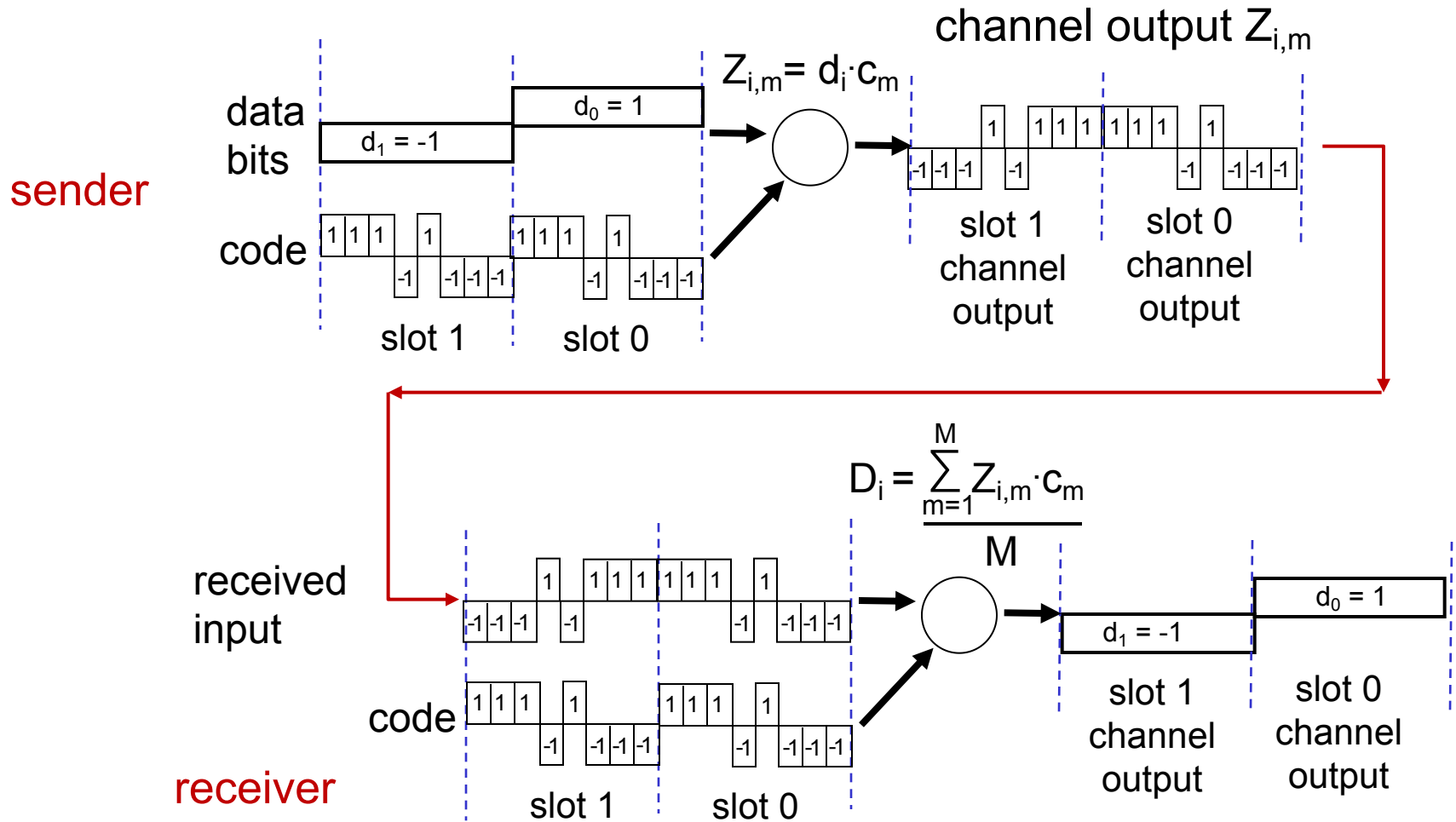
Fading:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other interfering at B

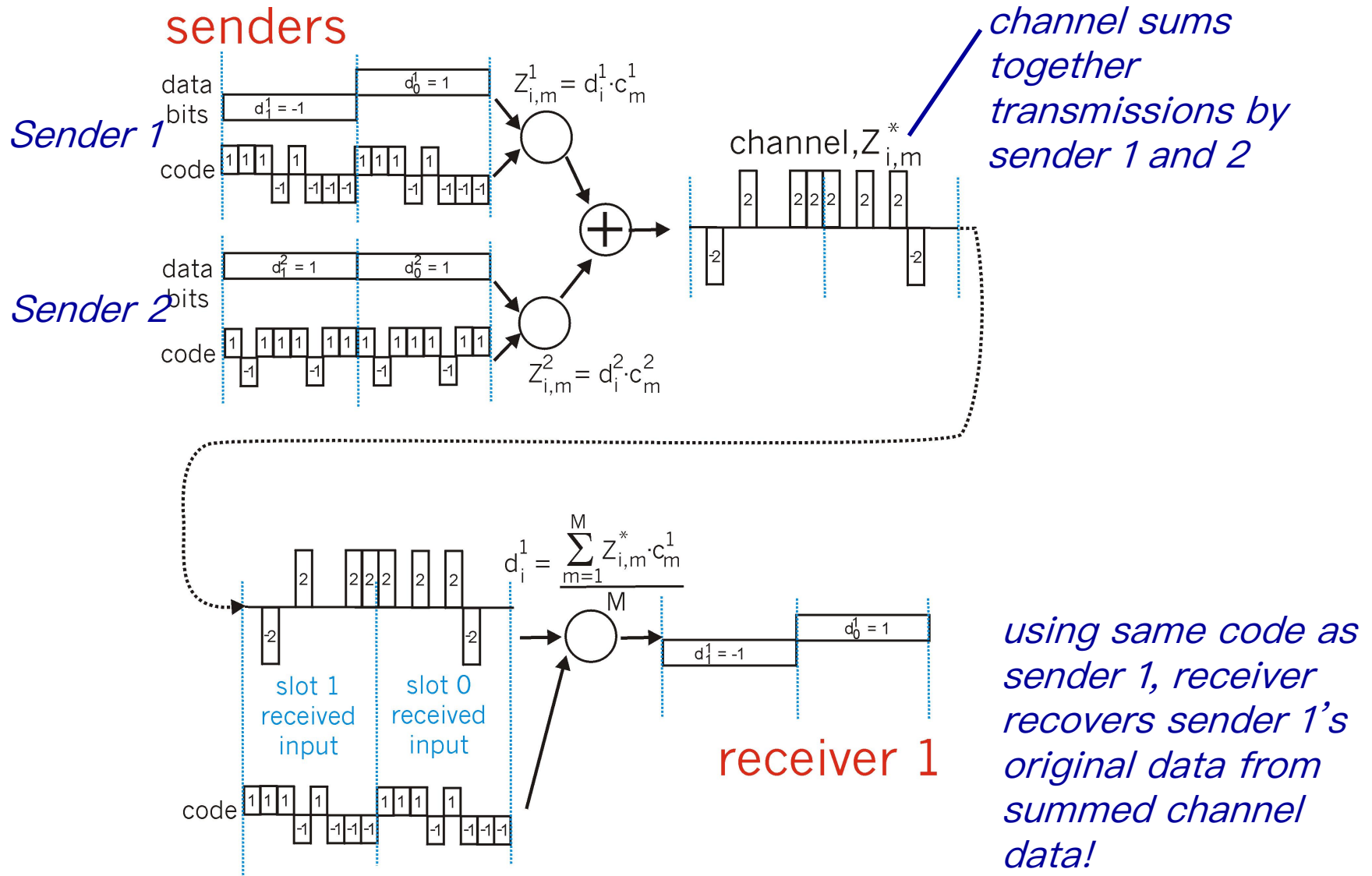
Code Division Multiple Access (CDMA)

- unique “code” assigned to each user; i.e., code set partitioning
 - all users share same frequency, but each user has own “chipping” sequence (i.e., code) to encode data
 - allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are “orthogonal”)
- *encoded signal* = (original data) X (chipping sequence)
- *decoding*: inner-product of encoded signal and chipping sequence

CDMA encode/decode



CDMA: two-sender interference



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IEEE 802.11 Wireless LAN

802.11b

- 2.4–5 GHz unlicensed spectrum
- up to 11 Mbps
- direct sequence spread spectrum (DSSS) in physical layer
 - all hosts use same chipping code

802.11a

- 5–6 GHz range
- up to 54 Mbps

802.11g

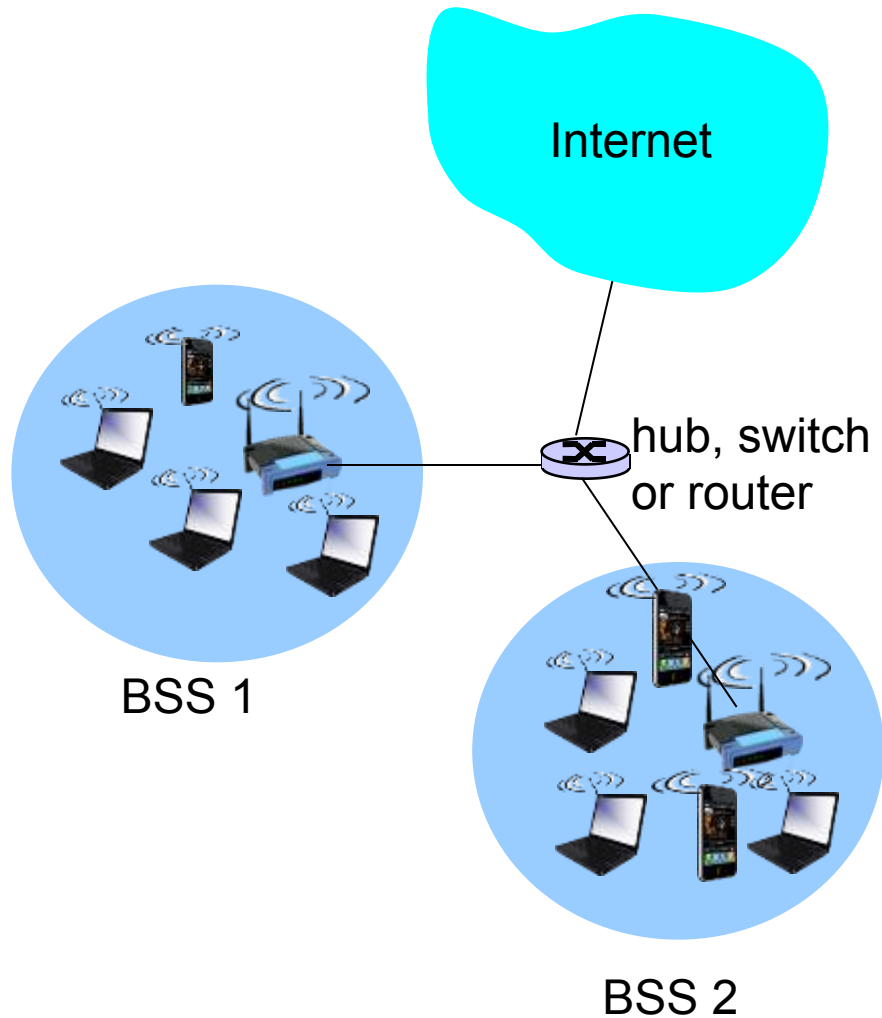
- 2.4–5 GHz range
- up to 54 Mbps

802.11n: multiple antennae

- 2.4–5 GHz range
- up to 200 Mbps

-
- all use CSMA/CA for multiple access
 - all have base-station and ad-hoc network versions

802.11 LAN architecture

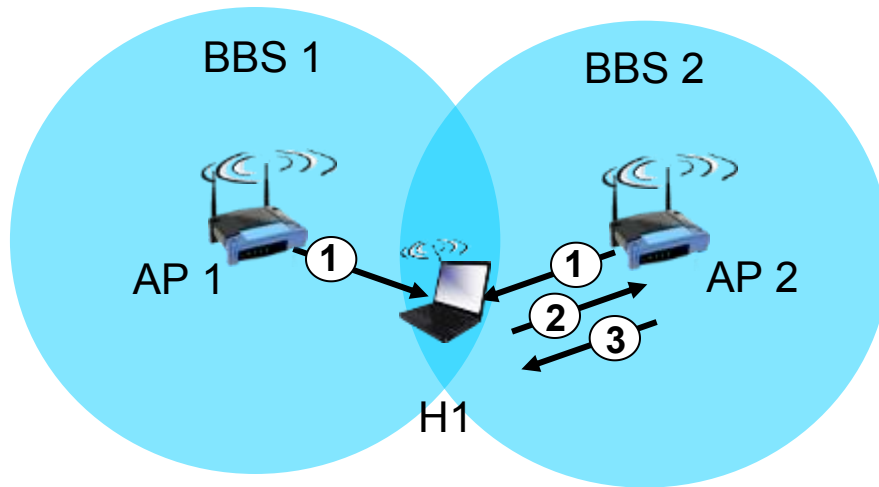


- wireless host communicates with base station
 - base station = access point (AP)
 - Basic Service Set (BSS) (aka “cell”) in infrastructure mode contains:
 - wireless hosts
 - access point (AP): base station
- ad hoc mode: hosts only

802.11: Channels, association

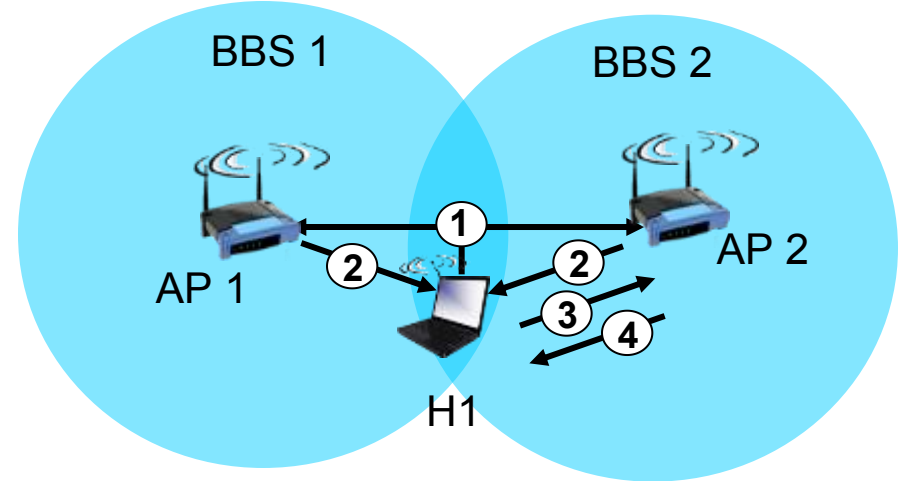
- 802.11b: 2.4GHz–2.485GHz spectrum divided into 11 channels at different frequencies
 - AP admin chooses frequency for AP
 - interference possible: channel can be same as that chosen by neighboring AP!
- host: must *associate* with an AP
 - scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
 - selects AP to associate with
 - may perform authentication [Chapter 8]
 - will typically run DHCP to get IP address in AP's subnet

802.11: passive/active scanning



passive scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent:
H1 to selected AP
- (3) association Response frame sent
from selected AP to H1

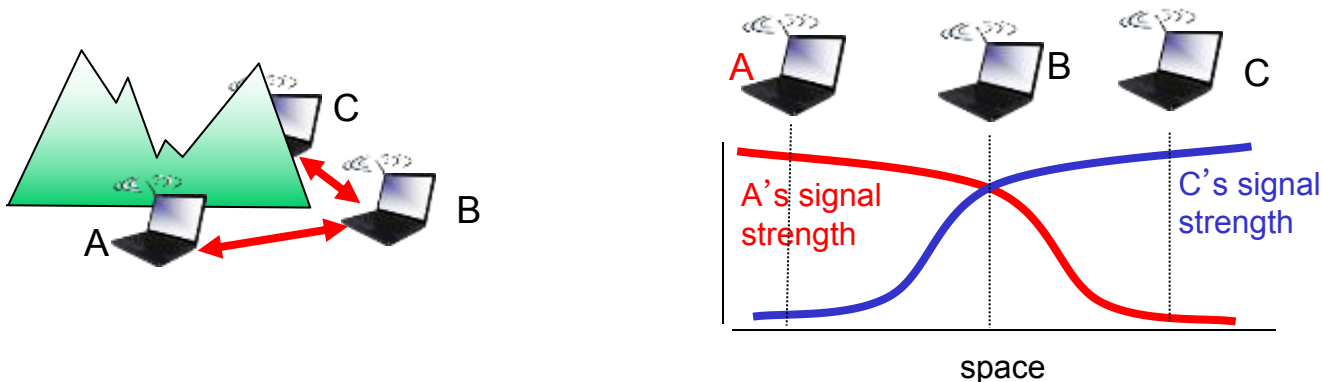


active scanning:

- (1) Probe Request frame broadcast
from H1
- (2) Probe Response frames sent
from APs
- (3) Association Request frame sent:
H1 to selected AP
- (4) Association Response frame sent
from selected AP to H1

IEEE 802.11: multiple access

- avoid collisions: 2+ nodes transmitting at same time
- 802.11: CSMA – sense before transmitting
 - don't collide with ongoing transmission by other node
- 802.11: *no* collision detection!
 - difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
 - can't sense all collisions in any case: hidden terminal, fading
 - goal: *avoid collisions*: CSMA/C(ollision)A(avoidance)



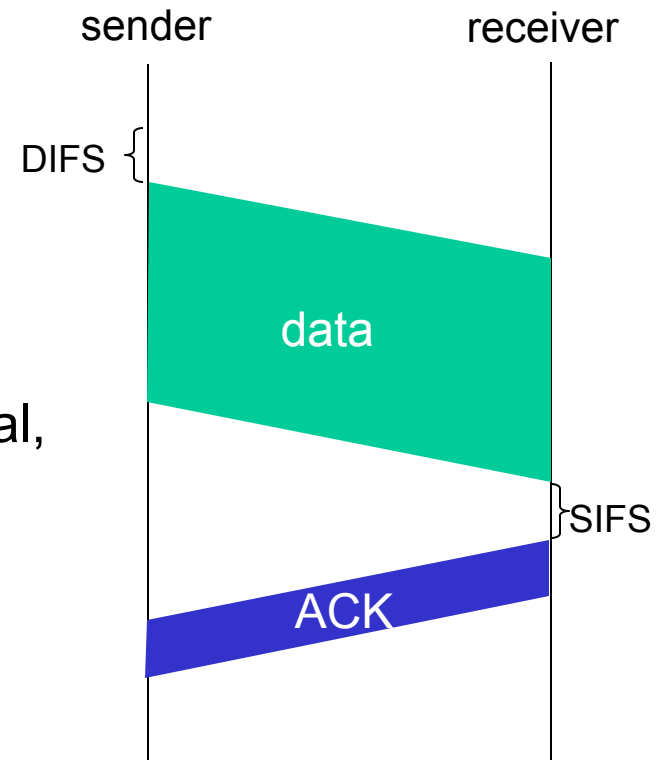
IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

- 1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)
- 2 if sense channel busy then
start random backoff time
timer counts down while channel idle
transmit when timer expires
if no ACK, increase random backoff interval,
repeat 2

802.11 receiver

- if frame received OK
return ACK after **SIFS** (ACK needed due to
hidden terminal problem)



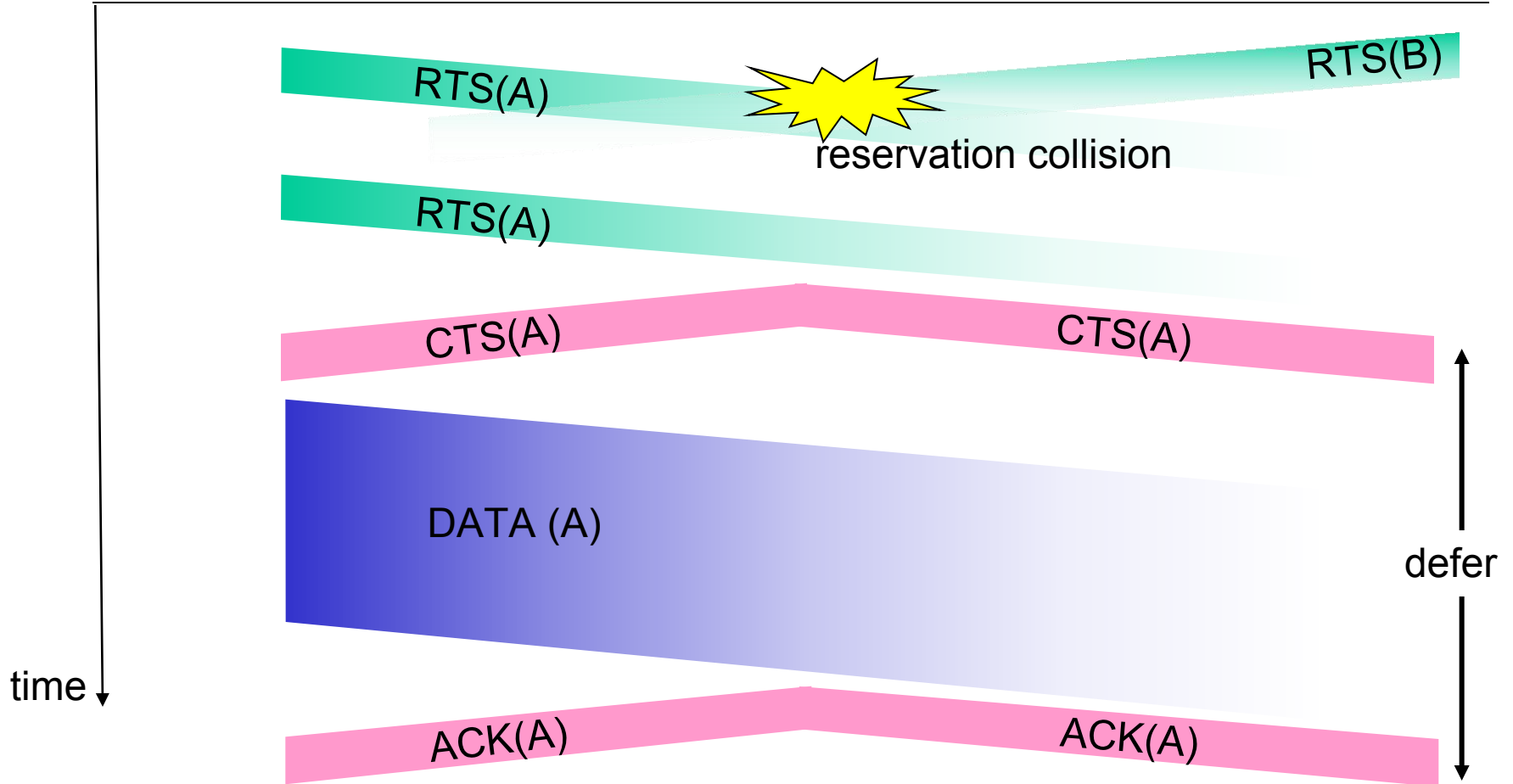
Avoiding collisions (more)

idea: allow sender to “reserve” channel rather than random access of data frames: avoid collisions of long data frames

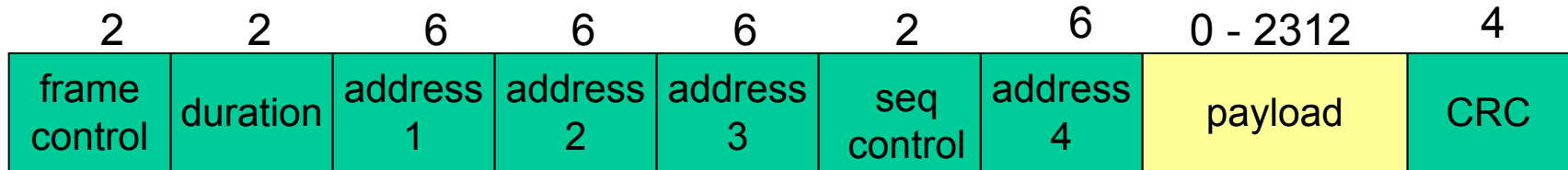
- sender first transmits *small* request-to-send (RTS) packets to BS using CSMA
 - RTSs may still collide with each other (but they’re short)
- BS broadcasts clear-to-send CTS in response to RTS
- CTS heard by all nodes
 - sender transmits data frame
 - other stations defer transmissions

*avoid data frame collisions completely
using small reservation packets!*

Collision Avoidance: RTS-CTS ~~exchange~~



802.11 frame: addressing



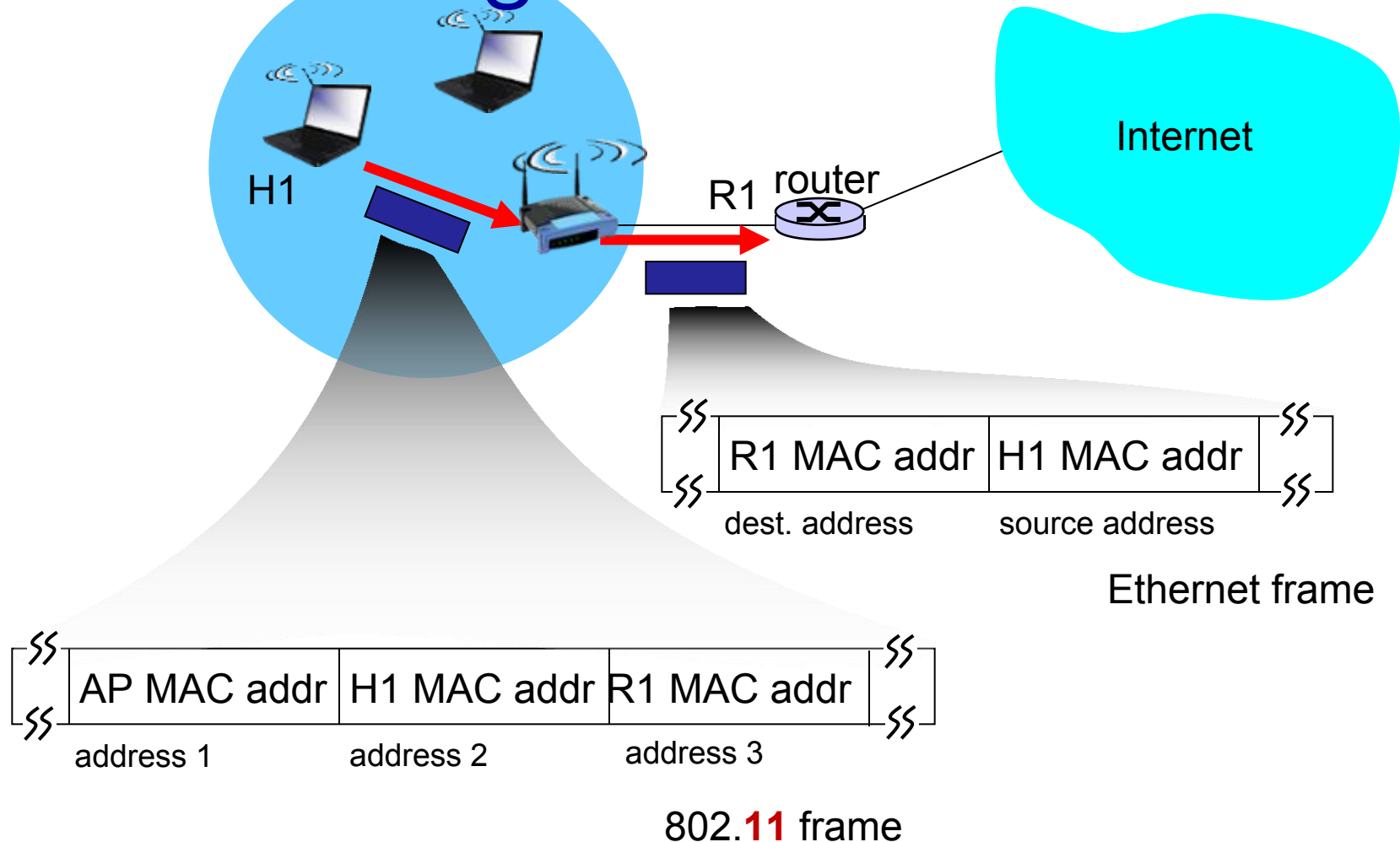
Address 1: MAC address of wireless host or AP to receive this frame

Address 2: MAC address of wireless host or AP transmitting this frame

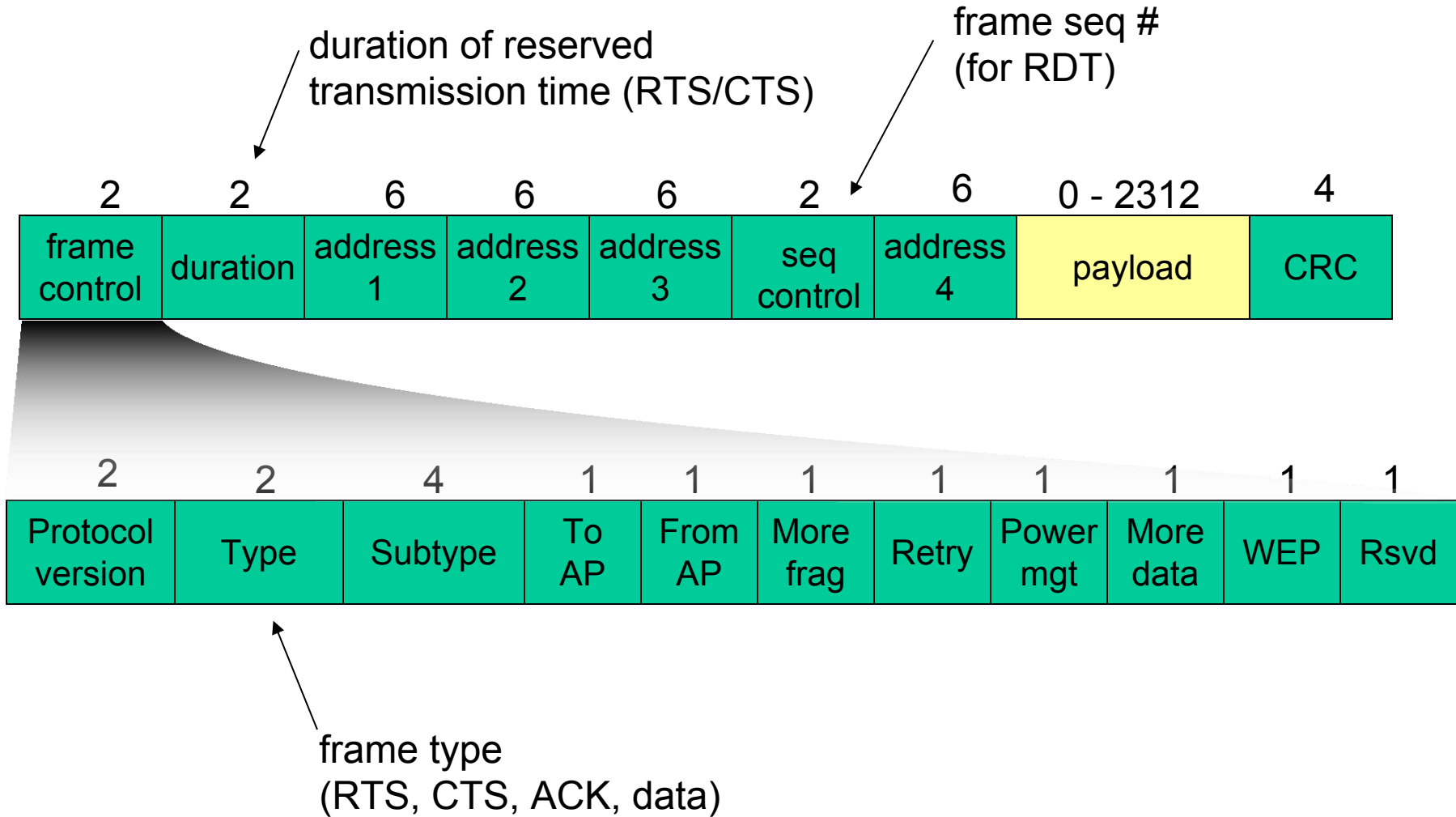
Address 3: MAC address of router interface to which AP is attached

Address 4: used only in ad hoc mode

802.11 frame: addressing

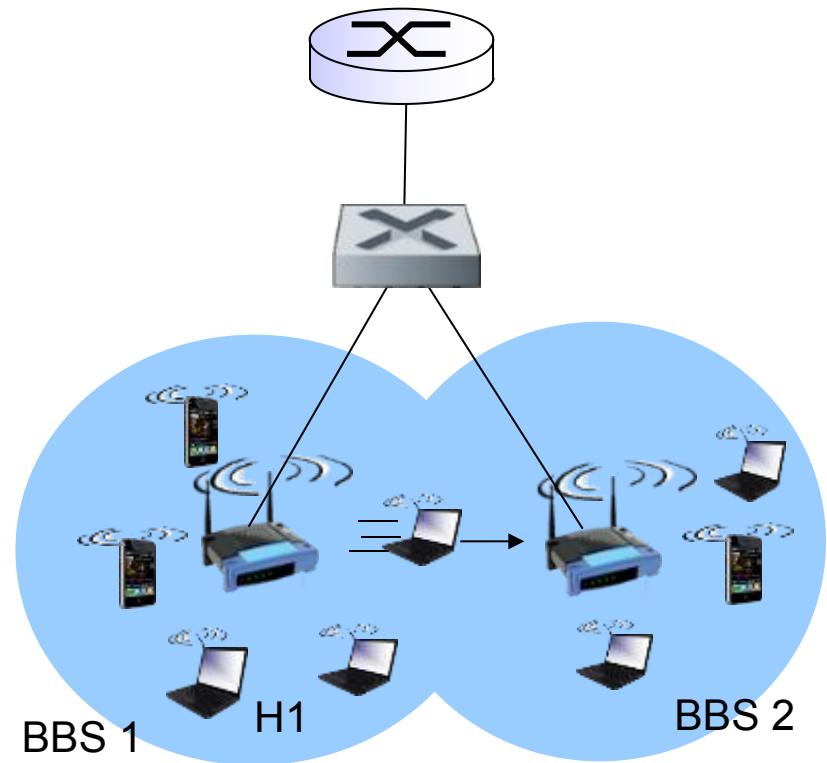


802.11 frame: more



802.11: mobility within same subnet

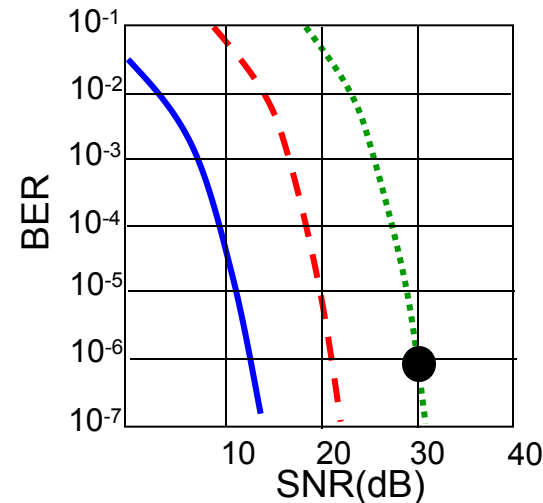
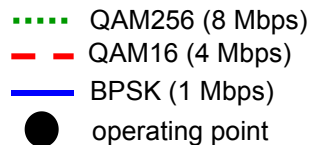
- H1 remains in same IP subnet: IP address can remain same
- switch: which AP is associated with H1?
 - self-learning (Ch. 5): switch will see frame from H1 and “remember” which switch port can be used to reach H1



802.11: advanced capabilities

Rate adaptation

- base station, mobile dynamically change transmission rate (physical layer modulation technique) as mobile moves, SNR varies



1. SNR decreases, BER increase as node moves away from base station
2. When BER becomes too high, switch to lower transmission rate but with lower BER

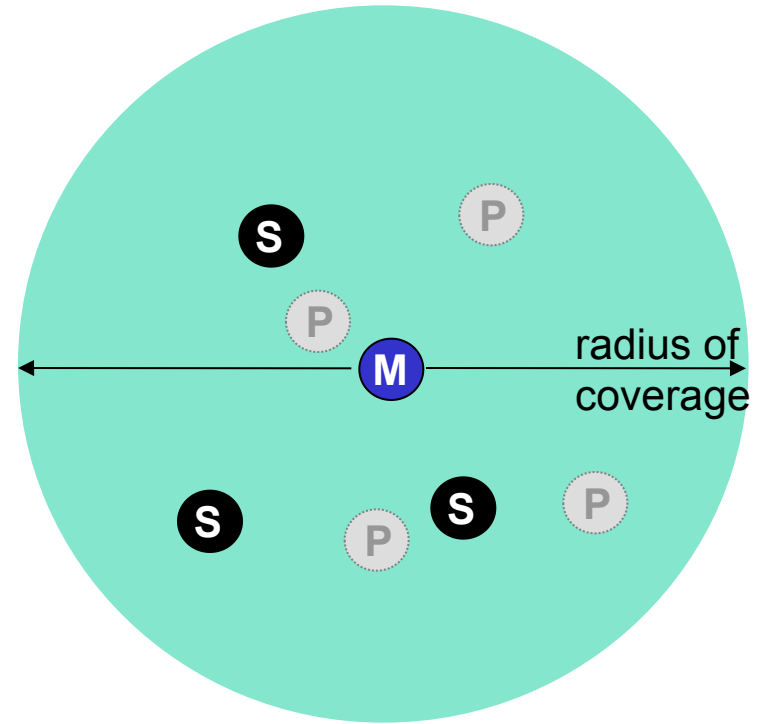
802.11: advanced capabilities

power management

- node-to-AP: “I am going to sleep until next beacon frame”
 - AP knows not to transmit frames to this node
 - node wakes up before next beacon frame
- beacon frame: contains list of mobiles with AP-to-mobile frames waiting to be sent
 - node will stay awake if AP-to-mobile frames to be sent; otherwise sleep again until next beacon frame

802.15: personal area network

- less than 10 m diameter
- replacement for cables (mouse, keyboard, headphones)
- ad hoc: no infrastructure
- master/slaves:
 - slaves request permission to send (to master)
 - master grants requests
- 802.15: evolved from Bluetooth specification
 - 2.4–2.5 GHz radio band
 - up to 721 kbps



- M** Master device
- S** Slave device
- P** Parked device (inactive)